The Nature for Cooling Rapid Assessment Tool

A transparent, evidence-grounded screening methodology for prioritising nature-based solutions for urban cooling

Methodology Report

Version 1 of the methodology, released for expert review

Criterra

https://criterra.eu

Methodology version	2026.08.02
Document date	30 July 2026
Status	Released for expert and public review
Licence	Apache License 2.0

Openness statement. The Nature for Cooling Rapid Assessment Tool, its complete methodology configuration, its source code, and this document are public. The tool is released under the Apache License, Version 2.0. Every quantitative value the tool applies is derived in this document, recorded in machine-readable configuration, and traced to a cited source whose verification status is disclosed in Appendix E. Readers are able to inspect, reproduce, contest, and modify every element of the methodology described here.

How to cite this document. Criterra (2026). *The Nature for Cooling Rapid Assessment Tool: a transparent, evidence-grounded screening methodology for prioritising nature-based solutions for urban cooling.* Methodology Report, version 2026.08.02. `https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool`

Corresponding methodology artefacts. The normative configuration files are `config/*.yaml`; the per-typology derivations are `docs/methodology/EVIDENCE-TABLES.md`; the source register with verification status is `docs/methodology/BIBLIOGRAPHY.md`. All carry the version stamp 2026.08.02 and move as a single unit.

Invitation to review, and how to reach us. This document is written to be contested. Reviewers are specifically invited to challenge the adopted cooling envelopes (Section 6), the aggregation weights (Section 7.11), the derivation of energy savings (Section 7.8), and the treatment of uncertainty (Section 8). Questions, corrections, and methodology challenges are handled openly as issues at `https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool/issues`, so that critique and its resolution remain part of the public record; Section 13 describes what a well-formed challenge looks like.

Abstract

Cities facing intensifying heat must decide where to intervene and with which nature-based solution, typically across many more candidate sites than can be studied in detail. The available options are unstructured judgment, which is fast but neither comparable nor defensible, and microclimate simulation, which is rigorous but too slow and costly for triage. This paper specifies a screening-level, multi-criteria methodology that occupies the gap between them. The methodology converts a structured description of an urban site and one proposed nature-based intervention into a Heat Priority Index, a Cooling Potential Score, an indicative daytime pedestrian-level air-temperature reduction range, derived cooling-energy and greenhouse-gas estimates, equity and co-benefit scores, and a single NbS Cooling Opportunity Score on a 0–100 scale, each accompanied by an explicit confidence rating. It is built on a library of fourteen intervention typologies whose performance envelopes are derived from published evidence, principally an umbrella review of sixty-four systematic reviews (Keravec-Balbot et al. 2026), an empirical meta-analysis of park cooling (Bowler et al. 2010), and a scale-dependent canopy study (Ziter et al. 2019); the composite-indicator construction follows established practice (Nardo et al. 2008) and the prioritisation logic follows Norton et al. (2015). Three calibration choices are emphasised: a downward correction for the modelling bias present in the synthesised literature, a refusal to claim super-additive cooling for combined intervention packages, and the clipping of all reported temperature ranges to the literature-supported envelope so that favourable site conditions can never generate a claim exceeding published evidence. The methodology's limitations are stated rather than minimised: all cooling values are daytime only; the quantity estimated is air temperature, not thermal comfort; the evidence base is geographically skewed towards temperate and subtropical Northern Hemisphere cities; no default costs are shipped; and the aggregation weights are declared expert judgment, including a disclosed equity-forward emphasis that gives vulnerability an effective weight of 0.25 against heat exposure at 0.15. The sensitivity analysis specified here is planned methodology, not a reported result. The tool, its configuration, and this document are public under the Apache License 2.0.

Keywords: urban heat island; nature-based solutions; urban cooling; composite indicator; screening assessment; heat vulnerability; decision support; evidence traceability.

Contents

1 Introduction

1.1 Urban heat as a decision problem

Urban populations are exposed to temperatures systematically higher than those of their rural surroundings, and the excess is not uniform across a city. The distinction between the surface, canopy-layer, and boundary-layer urban heat island, established by Oke (1976), remains the necessary starting point for any serious treatment of the subject, because the three are different physical quantities measured in different ways and they do not agree in magnitude. Within a single city, daytime near-surface air temperature can vary by several degrees over short distances: Ziter et al. (2019) measured a mean daytime intra-urban air temperature range of 3.5 °C (spanning 1.1–5.7 °C) and a mean nighttime range of 2.1 °C (1.2–3.0 °C) across a single urban area. Variation of this magnitude, over distances a person walks in minutes, is what makes urban heat a spatial planning problem rather than only a meteorological one.

The consequences of that variation are distributed unevenly across the population. Reid et al. (2009) reduced ten candidate vulnerability variables — six demographic, two describing household air-conditioning, one describing vegetation cover, and one describing diabetes prevalence — to four factors explaining more than 75 % of total variance: social and environmental vulnerability, social isolation, air-conditioning prevalence, and the proportion of elderly and diabetic residents. Two people experiencing the same air temperature do not face the same risk. Any instrument that ranks sites for intervention on physical heat alone is answering a narrower question than the one cities are actually asking.

1.2 Nature-based solutions as a cooling response

Vegetation and water cool urban environments through shade, evapotranspiration, and evaporation, and the empirical evidence for the effect is well established at the scale of individual green sites. Bowler et al. (2010), restricting their meta-analysis to empirical studies, found that an urban park was on average 0.94 °C cooler during the day than its non-green surroundings, with larger parks and parks containing trees cooler than smaller or open ones. Ziter et al. (2019) showed that the relationship between canopy cover and daytime air temperature is nonlinear, with the greatest cooling occurring above approximately 40 % canopy cover, and that the effect is scale-dependent, peaking at radii of 60–90 m — roughly the scale of a city block. The same study found that impervious cover raises temperature approximately linearly and that its warming effect persists overnight, where canopy cooling largely does not.

Aggregating across measures and climates, Keravec-Balbot et al. (2026) conducted an umbrella review of 64 systematic reviews with a multilevel meta-regression of pedestrian-level daytime air temperature reductions. That synthesis is the single most useful cross-typology anchor currently available, and it carries a finding with direct consequences for tool design: the type of intervention alone explains only 12 % of the variance in observed cooling effectiveness, rising to 30 % when climate zone is added and to 78 % once the Köppen–Geiger climate subzone is included. An instrument that reports one global cooling figure per intervention type is therefore discarding the majority of the explanatory signal in the evidence base.

1.3 The decision gap

A city preparing a heat adaptation programme must choose among many more candidate sites and interventions than it can study in detail. In practice it has two options, and neither serves the triage step.

The first is unstructured judgment, often mediated by a spreadsheet. It is fast and cheap, and it draws on genuine local expertise. Its weakness is not that it is wrong but that it is not *comparable*: two officers assessing two sites apply different implicit weights, cite no evidence that a third party can check, and produce numbers that cannot be defended to a funder or a public inquiry. Nothing about the process records why one site outranked another.

The second is detailed simulation — microclimate models of the ENVI-met class, computational fluid dynamics, or building energy simulation. These are the appropriate instruments for design, and they are rigorous. The work of Zölch et al. (2016) illustrates both their value and their cost: a micro-scale ENVI-met evaluation of heat mitigation measures which established, among other things, that trees achieved a mean physiologically equivalent temperature (PET) reduction of 13 % at pedestrian level, that green façades achieved 5–10 %, that green roof effects at pedestrian level were negligible, and — a finding of general importance — that increasing green cover did not correspond linearly to PET reduction, because placement matters more than quantity. Producing such results requires geometry, material properties, meteorological forcing, and specialist time. A city cannot run them across two hundred candidate street segments before deciding which twenty to investigate.

The gap between the two is the screening step: a structured, repeatable, evidence-referenced first pass that narrows a long list to a short one and records why. That is the step this methodology addresses.

1.4 Contribution and structure of this paper

This paper specifies, in full, the methodology of the Nature for Cooling Rapid Assessment Tool at methodology version 2026.08.02. It is written to be read independently of the software, by reviewers who wish to evaluate the scientific basis of the tool's outputs before deciding whether to trust, adopt, or contest them. Its contributions are four.

1. **A complete, auditable specification.** Every formula the engine applies is stated and typeset here, every weight is declared with its justification or with an explicit statement that it is a value choice, and every performance value is traced to a cited source whose verification status is disclosed (Appendix E).
2. **An evidence-derived typology library.** Fourteen intervention typologies carry cooling envelopes derived from published literature, with the derivations exposed rather than summarised. Three calibration decisions that materially lower the tool's headline numbers — a downward correction for modelling bias, the refusal of super-additivity for combined packages, and conservative treatment where sources conflict — are presented for scrutiny in Section 6.5.
3. **A discipline of honest uncertainty.** All quantitative outputs are ranges; every output block carries its own confidence rating; missing data reduces confidence rather than being silently

imputed; and the temperature-envelope clipping rule prevents favourable site conditions from generating claims that exceed published evidence (Section 7.6).

4. **Reproducibility as an engineering property.** The methodology lives in versioned configuration files rather than in code, and its evidence rules are enforced by the build: an uncited performance value or a citation pointing at an unrecorded source fails continuous integration (Section 10). This is what allows the claims in this paper to be checked rather than merely believed.

Section 2 states what the tool is and is not, and the design commitments that constrain it. Section 3 positions it against published work and argues for the standards by which a screening instrument should be judged. Section 4 presents the three-layer conceptual structure. Sections 5 to 7 specify the inputs, the typology library, and every scoring formula. Sections 8 and 9 cover uncertainty treatment and the planned sensitivity analysis. Section 10 describes the implementation and the machine-enforced evidence rules. Section 11 works one illustrative assessment through by hand. Sections 12 and 13 state the limitations and the governance process. Appendices reproduce the full typology library, the complete weight tables, the input field reference, a glossary, and the source verification register.

2 Scope and intended use

2.1 What the tool is

The Nature for Cooling Rapid Assessment Tool is a screening-level, multi-criteria decision-support instrument. It converts a structured description of an urban site and one proposed nature-based intervention into comparable scores and indicative quantitative estimates, so that early planning and investment decisions can be made on a transparent, evidence-referenced basis.

A user describes a site through six input groups, selects one of fourteen intervention typologies, and receives a Heat Priority Index, a Cooling Potential Score with an indicative daytime pedestrian-level air temperature reduction range, derived cooling-energy and greenhouse-gas estimates, cost outputs where cost data has been supplied, equity and co-benefit scores, a single NbS Cooling Opportunity Score for comparing options, per-block confidence ratings, and an itemised list of every default the engine applied. Interventions for the same site can be assessed and compared side by side.

2.2 What the tool is not

The following exclusions are definitional, not caveats to be read past.

- **It is not a microclimate simulation.** It does not solve an energy balance, does not represent site geometry, and does not predict the temperature at a specific location at a specific time. Its outputs are not comparable to ENVI-met-class or CFD results and must never be presented as such.
- **It is not a building energy model.** Cooling-energy savings are derived from an estimated air temperature reduction through a published temperature–electricity-demand sensitivity

(Section 7.8). No building is modelled.

- **It is not a planting design or species selection tool.** The evidence that species and leaf area index materially change delivered cooling (Rahman et al. 2020) is used only to justify that canopy condition modulates performance. The tool gives no species guidance.
- **It is not a regulatory, engineering, or approval instrument.** Outputs are indicative ranges for comparison and prioritisation. They are not design specifications and they are not contractual performance guarantees.
- **It does not replace site investigation or local expertise.** It structures a judgment; it does not substitute for one.

2.3 Intended users and decisions

Three user groups are supported: municipal planners and public agencies; developers and consultants; and researchers and non-governmental organisations. The tool is designed to inform three questions:

1. Does this site merit attention at all, given how hot it is and who is exposed?
2. Which of several candidate interventions best suits this particular site?
3. How does a proposed intervention perform simultaneously across cooling, energy, climate, cost, equity, and co-benefit dimensions?

Each of these is a comparative question. The tool is built to rank and to compare, and its outputs should be interpreted comparatively. An Opportunity Score of 72 is meaningful principally in relation to the score another option or another site receives under the same methodology version; it is not a physical measurement.

2.4 The five design commitments

Five commitments constrain the methodology. They are listed here because several of them cost the tool apparent capability, and a reader should understand that those costs were chosen.

Table 1: Design commitments and their consequences for the methodology.

Commitment	**Consequence**
Transparency	Every score has exactly one documented formula, reproduced in this paper. No machine learning, no opaque calibration, no undisclosed adjustment. A user who disagrees with a number can locate the rule that produced it.
Determinism	Identical inputs under an identical methodology version always yield identical outputs. There are no stochastic components anywhere in the scoring or reporting path, including the recommendation text, which is composed from templates rather than generated.
Evidence traceability	Every performance value cites a source. Values that could not be grounded in retrieved literature are either omitted — as unit costs are (Section 7.10) — or shipped with an explicit low-confidence flag.
Honest uncertainty	All quantitative outputs are ranges. Every result carries a confidence rating, computed separately per output block. Missing data reduces confidence rather than being silently imputed, and every applied default is itemised in the result.
Auditability	The methodology lives in versioned configuration files, not in code. Every result records the methodology version that produced it, and the evidence rules are enforced by continuous integration rather than by review convention (Section 10).

The commitment to honest uncertainty is the one that most visibly reduces the tool's apparent authority. A screening instrument that reported single numbers would look more decisive and would be easier to put in a slide. The adopted cooling envelopes span factors of three to ten, and the climate-differentiated estimates in Keravec-Balbot et al. (2026) carry uncertainty intervals that in several cases approach or exceed ± 1 °C, and in the case of street trees in arid climates reach ± 1.4 °C on a central estimate of 1.7 °C; a point estimate would imply a precision the evidence does not support. The ranges are the honest representation of what is known.

3 Related work and positioning

3.1 The closest published antecedent

Norton et al. (2015) is the closest published antecedent to this methodology. It sets out a framework for prioritising and selecting urban green infrastructure to mitigate high temperatures, built on a review of the relationships between urban geometry, green infrastructure, and temperature. Its logic — assess the thermal condition of a place, consider who is exposed, then determine which intervention is suited to that place — is the logic this tool implements, and the three-layer structure of Section 4 follows it directly.

The verification status of this source should be stated plainly, because the paper leans on it for framing. The bibliographic record was confirmed from the Monash University research portal, and the framework description was confirmed from an authoritative secondary description rather than from the full text. The tool takes no numeric value from Norton et al. (2015); it takes a structural argument. Appendix E records this alongside every other source.

Two differences from that antecedent are worth stating. First, this tool is an operational instrument with a fixed, versioned, machine-readable parameterisation rather than a conceptual framework, which means every judgment it embodies must be written down as a number and can therefore be disputed as a number. Second, it treats climate as a first-order term rather than as context, for the empirical reason given in Section 3.4.

3.2 Composite-indicator practice

The tool is, structurally, a composite indicator: a set of normalised sub-indicators combined by weighted aggregation into a headline score. The construction sequence adopted here — theoretical framework, variable selection, treatment of missing data, normalisation, weighting and aggregation, and presentation — follows Nardo et al. (2008), the standard reference for this class of instrument.

Two of that handbook's positions are load-bearing for this paper and are adopted without qualification.

The first is that **weighting is a value choice, not a technical derivation**. No literature can supply the correct relative importance of cooling performance against equity against cost, because that ratio expresses a policy preference rather than a fact about the world. The aggregation weights in Section 7.11 are therefore declared as expert judgment, versioned, adjustable in configuration, and are to be defended empirically through the sensitivity analysis specified in Section 9 rather than by appeal to authority. Presenting them as derived would be the more persuasive move and the dishonest one.

The second is that **uncertainty and sensitivity analysis are integral to a credible composite indicator, not optional additions**. Because the aggregation weights are the tool's most consequential unsourced choice, their influence on the output must be quantified and published rather than asserted to be small. Section 9 specifies that analysis in full and Section 9.4 reports its results at version 2026.08.02.

3.3 The risk framing

The Baseline layer follows the standard climate-risk decomposition, in which risk arises from the interaction of a *hazard* with *exposure* and *vulnerability*. This is why a physically hot site with no exposed population scores differently from an equally hot site surrounded by vulnerable residents, and it is the reason the tool computes a Heat Priority Index rather than reporting heat alone.

The decomposition is applied concretely: the Heat Exposure Score (Section 7.2) represents hazard and physical exposure, drawing on surface-energy-balance drivers for which Ziter et al. (2019) supplies direct empirical support; the Vulnerability Score (Section 7.3) represents population susceptibility, with its indicator selection grounded in Reid et al. (2009); and the two are combined into a single priority index (Section 7.4).

3.4 Climate as a first-order term

The tool's most consequential departure from common screening practice is its refusal to publish a single global cooling number per intervention type. The justification is empirical. Keravec-Balbot et al. (2026) reports that the type of solution alone explains 12 % of the variance in observed cooling effectiveness; adding climate zone raises this to 30 %; and adding the Köppen–Geiger climate subzone raises it to 78 %.

Figure 1 shows what that variance decomposition looks like at the level of individual measures. For green walls, the same synthesis that reports a 3.0 °C global central estimate reports subzone-specific estimates ranging from (5.7 ± 1.0) °C in Cwa and (5.5 ± 1.0) °C in BWh down to 1.3 °C in Cfa/Cfb and (1.0 ± 0.8) °C in Af — an approximately fivefold spread within a single measure. For street trees, the same review reports (4.0 ± 1.6) °C in tropical (A) climates against (1.7 ± 1.4) °C in arid (B) climates.

A tool reporting one number per measure would discard this. Climate suitability is consequently a first-order adjustment in the tool's Layer 2 (Section 7.5), not a refinement applied afterwards.

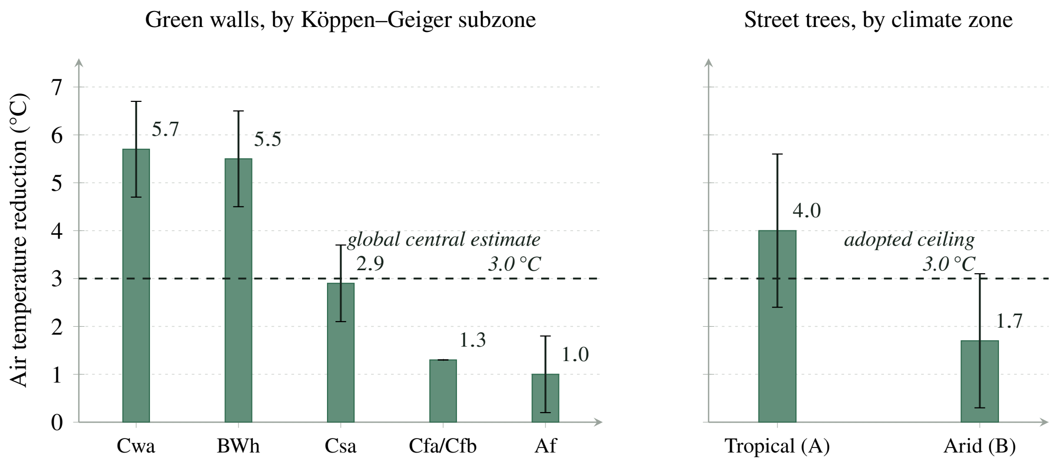

Figure 1: Climate dependence of reported cooling effectiveness, from the umbrella review of Keravec-Balbot et al. (2026). All values are daytime pedestrian-level *air* temperature reductions. Error bars show the uncertainty the source states; no bar is drawn for Cfa/Cfb, which this methodology's evidence tables record as a single combined central estimate without a stated interval. The dashed line marks 3.0 °C: the global central estimate for green walls in the left panel, and the ceiling this methodology adopts for any single site-level intervention in the right panel. The spread within each measure is the empirical reason climate suitability is treated as a first-order adjustment rather than as context.

3.5 By what standard should a screening tool be judged?

A recurring failure in the reception of screening instruments is that they are evaluated against simulation standards, which they cannot meet and were not built to meet. If the question is "does this reproduce ENVI-met to within 0.2 °C?", the answer is no, and it would be no for any instrument that asks forty-five questions and runs in seconds. That is not a finding about this tool; it is a restatement of what a screening instrument is.

The productive questions are different, and this paper invites reviewers to apply them.

1. **Is every value traceable?** Can a reader locate the source of every number the tool applies, and see what was actually verified about that source? Appendices A and E exist to make this checkable.
2. **Are the uncertainties honestly represented?** Does the tool report ranges that reflect the spread in the underlying evidence, and does it lower its stated confidence when it knows less? Section 8 specifies the mechanism.
3. **Is the ranking stable?** Because prioritisation is the tool's actual function, the relevant question about the weights is not whether they are correct — they are a value choice — but whether the ordering they produce survives reasonable perturbation. Section 9 specifies how this will be measured and published.
4. **Are the value judgments disclosed?** Where the methodology takes a position that a reasonable person could reject — most notably the equity-forward weighting of Section 7.12 — is that position stated openly and quantified, or is it buried in an aggregation?
5. **Does it refuse to answer when it should?** An instrument that always produces a number is less trustworthy than one that reports *not estimated* when its inputs do not support an estimate. The cost policy of Section 7.10 and the missing-data rules of Section 8.4 are deliberate refusals.

A screening tool that satisfies these five is doing its job. One that fails them is not rescued by agreeing with a simulation on a test case.

4 Conceptual framework

4.1 Three layers, three questions

The methodology is organised into three layers, each answering a distinct question that a planner asks in sequence. The structure is shown in Figure 2.

Layer 1 — Baseline assessment: ***does this site deserve attention?***
A Heat Exposure Score and a Vulnerability Score are computed from the site description and combined into a Heat Priority Index. This layer is independent of the proposed intervention: it characterises the place, not the project. Two options assessed for the same site share an identical Layer 1 result, which is what makes side-by-side comparison meaningful.

Layer 2 — NbS performance: ***what can this intervention deliver here?***
A base value drawn from the typology library is modulated by a site adjustment factor derived from conditions the user has already described. This layer produces the Cooling Potential Score and the indicative temperature reduction range. It is where the tool's evidence base does its work, and where the clipping rule of Section 7.6 constrains what may be claimed.

Layer 3 — Impact and feasibility: ***what does it cost, and what else does it do?***
Cooling-energy savings are derived from the Layer 2 temperature estimate and converted to avoided greenhouse-gas emissions; capital cost, where supplied, yields payback and a derived cost-feasibility score; and co-benefit and equity scores capture what a purely thermal assessment would miss.

The three layers are then aggregated into a single NbS Cooling Opportunity Score, accompanied by per-block confidence ratings, any suitability or evidence flags, and a deterministically composed recommendation.

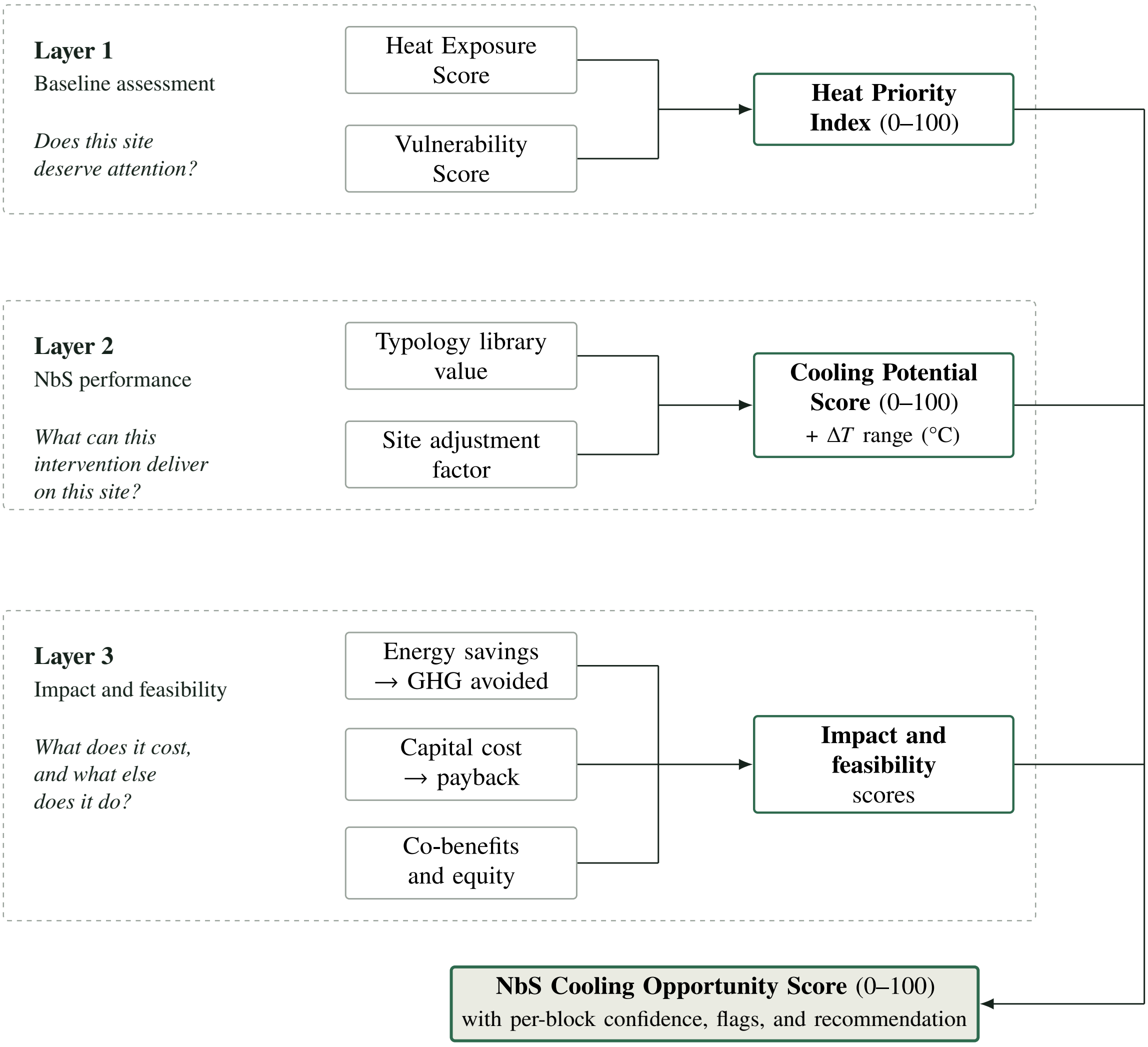

Figure 2: The three-layer conceptual framework. Layer 1 characterises the site independently of any intervention; Layer 2 evaluates one proposed intervention against that site; Layer 3 adds economic, climate, and social dimensions. The layers are aggregated by the weighted sum of Section 7.11. Because Layer 1 does not depend on the intervention, two options assessed for the same site differ only from Layer 2 onward.

4.2 Why this decomposition

The separation is not merely presentational; it has three consequences that matter for how outputs should be read.

It separates the place from the project. A high Heat Priority Index with a low Cooling Potential Score identifies a site that badly needs intervention but for which the proposed intervention is poorly matched — a result that a single blended score would conceal. This pattern is common and decision-relevant: it argues for a different intervention, not for abandoning the site.

It keeps the evidence base localised. Layer 2 is where published cooling evidence enters, and it enters in one place: the typology library of Section 6. Revising the evidence base means revising that library and its adjustment rules, not tracing consequences through the whole methodology. This is what makes the version-controlled configuration described in Section 10 tractable.

It makes the confidence structure natural. A site typically has good physical data and poor economic data, or the reverse. Because the layers are distinct, confidence can be computed and reported per block (Section 8.3) rather than collapsed into a single figure that would be dominated by whichever block happened to be worst.

4.3 From inputs to outputs

Figure 3 shows the same structure as an implementation data flow. Three boundaries in that diagram carry the design commitments of Section 2.4 into the software.

First, methodology values reach the engine only as configuration. The scoring code contains no typology value, no weight, and no threshold; all of these live in `config/*.yaml` and are validated on load. Changing the methodology never requires changing code, and the methodology version travels with the result.

Second, the engine is a pure function of its inputs and that configuration. It performs no input or output, makes no network call, holds no global state, and contains no stochastic component. Determinism is therefore a structural property rather than a testing aspiration.

Third, every output is a range or a categorical judgment accompanied by its confidence, and every default the engine applied is itemised in the result. A reader of a report can see exactly which numbers came from the user and which from the tool.

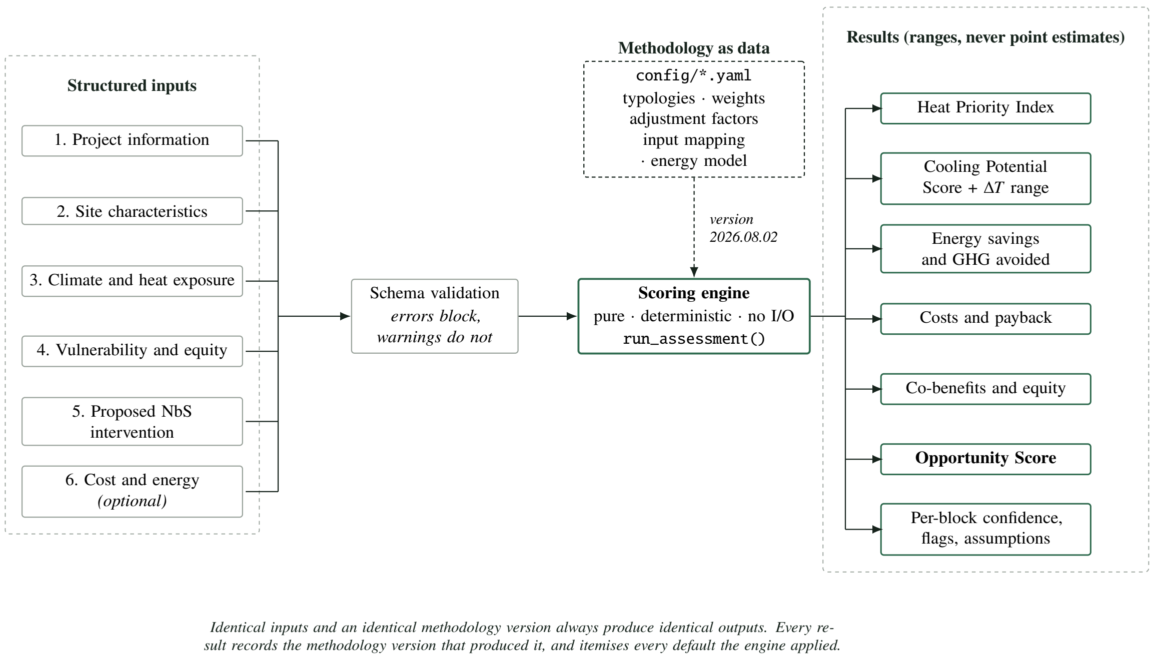

Figure 3: Data flow from the six input groups through validation and the scoring engine to the result. Methodology values enter only as versioned configuration (dashed path); the engine is a pure, deterministic function of inputs and configuration; and every output is reported as a range or a categorical judgment with its own confidence rating. Validation distinguishes errors, which block an assessment, from warnings, which do not.

5 Inputs and normalisation

5.1 Input structure

Inputs are collected in six groups: project information; site characteristics; climate and heat exposure; vulnerability and equity; the proposed NbS intervention; and cost and energy parameters. The full field reference, with required or optional status and the normalisation applied to each field, is reproduced in Appendix C.

Every input is either *required*, in which case the assessment cannot proceed without it, or *optional*, in which case its absence reduces confidence but never blocks the assessment. The interaction design anticipates roughly forty-five inputs, of which it expects about twenty to be mandatory; the implemented input schema requires exactly the four fields that a documented formula cannot be evaluated without, and Appendix C records the status and normalisation of every field at version 2026.08.02. Reducing the total count would not reduce uncertainty; it would move uncertainty from visible — a stated confidence level — to invisible — a silent default. The interface consequently asks everything, makes skipping free, and shows the user what skipping costs.

5.2 Qualitative normalisation

Most inputs accept a qualitative level, because most users at screening stage do not have measured data. Levels map to a 0–100 scale as shown in Table 2.

Table 2: Normalisation of qualitative input levels. The standard scale applies where a higher level indicates higher risk or higher quantity; the inverted scale applies where a higher level indicates *lower* risk.

Level	Standard scale	Inverted scale
Low	25	100
Medium	50	50
High	75	25
Very high	100	—
Unknown	50	50

Three fields use scales of their own, because their natural vocabularies are not the four standard levels. Soil availability maps *none* → 25, *limited* → 50, *moderate* → 75, *high* → 100. Irrigation availability maps *none* → 25, *occasional* → 50, *reliable* → 100. Current shade level maps *very low* → 25, *low* → 50, *medium* → 75, *high* → 100. In each case *unknown* maps to 50.

5.3 The inverted mapping for cooling access

The principal application of the inverted scale is access to cooled indoor space, where *high* access maps to 25, *medium* to 50, and *low* to 100. The field enters the Vulnerability Score (Section 7.3) as a *cooling access deficit*: low access indicates higher vulnerability, so it must produce a higher score.

The justification is Reid et al. (2009). In that factor analysis of ten heat vulnerability variables, air-conditioning prevalence emerged as one of four factors explaining more than 75 % of total variance — and, importantly, as an *independent* dimension. That independence is what justifies giving cooling access its own term in the vulnerability formula instead of allowing it to be absorbed into a general deprivation indicator: a factor that emerges separately alongside the social and environmental vulnerability factor is not a restatement of it. That last step is this methodology's inference from what a factor analysis means, not a claim the source itself makes.

Two qualifications should be read alongside this. The study population was North American, and access to mechanical cooling has a different distribution and a different meaning in cities where air conditioning is rare across all income groups. And the tool asks about access to cooled indoor space generally rather than about air-conditioning ownership specifically, which is a deliberate broadening — a shaded, ventilated, publicly accessible cool refuge counts — but a broadening that the underlying study does not itself validate.

5.4 The treatment of *unknown*

Assigning *unknown* a neutral score of 50 is a deliberate and disclosed choice. Three alternatives were available and each carries a bias.

Treating an unknown as the best case flatters sites with poor data; treating it as the worst case penalises them; and imputing a value from other fields introduces a correlation the user never asserted and cannot see. The neutral midpoint avoids both optimistic and pessimistic bias in the score, at the cost of pulling every score toward the middle in proportion to how much is unknown.

That cost is not concealed. The confidence mechanism of Section 8.3 records, per output block, how much of the relevant input set was actually supplied, so a score dominated by neutral defaults is reported at low confidence. The two mechanisms are complementary: the score says what the evidence supports, and the confidence says how much evidence there was. Neither alone is sufficient, and quoting a score without its confidence rating is identified in Section 12.9 as a misuse of the tool.

5.5 Land surface temperature as an optional proxy

Where a user can supply a land-surface-temperature (LST) anomaly — degrees Celsius above the city mean, typically derived from satellite imagery — the tool converts it linearly to a 0–100 score, with 0 °C mapping to 0 and 10 °C mapping to 100, clamped at both ends:

$$S_{\mathrm{LST}} = \mathrm{clamp}(10 \times a_{\mathrm{LST}})\,, \tag{1}$$

where a_{LST} is the supplied anomaly in degrees Celsius and clamp bounds its argument to $[0, 100]$. The 10 °C ceiling is consistent with observed intra-urban surface anomaly ranges while keeping the normalisation interpretable.

This proxy is used under a stated restriction, and the restriction is the most important sentence in this subsection. Land surface temperature is not air temperature, and the two diverge by a large and systematic factor.

Venter et al. (2021) contrasted crowdsourced air temperature measurements against satellite estimates and found the satellite-derived *surface* urban heat island to substantially exceed the air-temperature (*canopy-layer*) urban heat island: a reported mean surface UHI of 1.45 °C against a canopy-layer UHI of 0.26 °C, an approximately sixfold difference. This reflects the physical distinction between surface, canopy-layer, and boundary-layer heat islands established by Oke (1976).

The tool therefore treats LST **only as a relative indicator of which parts of a city are hotter than others**. It is never interpreted as an air-temperature value, never converted into a °C outcome, and never compared against the tool's own air-temperature reduction estimates. It influences only the ranking of sites within the Heat Exposure Score.

The verification status of Venter et al. (2021) is relevant here: the bibliographic record was confirmed from the ADS record and an author-hosted PDF, and the specific surface-versus-canopy comparison was confirmed from an indexed secondary description, publisher full text being blocked. The claim the methodology rests on — that surface UHI substantially exceeds canopy-layer UHI — is corroborated conceptually by Oke (1976), and the tool's use of the finding is qualitative (treat LST as relative only) rather than numerical, so no tool output depends on the precise sixfold ratio.

5.6 Where the tool has no LST

Most screening-stage users will not have an LST anomaly. The methodology therefore provides a second path through the Heat Exposure Score (Section 7.2) that requires only a qualitative heat exposure level. The two paths are not equivalent in information content, and the confidence mechanism reflects that. The data-poor path is not a degraded mode to be avoided; it is the expected mode for most assessments, and the tool is designed to give a defensible answer in it.

6 The NbS typology library

6.1 Structure

Fourteen intervention typologies are supported, in four families:

Green street tree planting, shaded pedestrian corridor, pocket park, urban forest / dense canopy planting, park upgrade, schoolyard greening.

Building green roof, green façade.

Blue-green rain garden / bioswale, blue-green corridor, riparian restoration.

Hybrid permeable shaded plaza, courtyard greening, mixed NbS package.

Each typology carries a base cooling score on a 0–100 scale, a temperature reduction envelope expressed as a minimum and a maximum in degrees Celsius, its primary cooling mechanism, a set of suitability conditions, an evidence confidence rating, default co-benefit levels, and its source citations. The complete library, including every citation and the specific finding each citation supports, is reproduced in Appendix A.

6.2 The metric, stated once and enforced throughout

> All adopted temperature values in this library are **daytime, pedestrian-level *air* temperature reductions**, relative to an equivalent unimproved site.
>
> They are **not** land surface temperatures, and they are **not** thermal comfort indices such as the physiologically equivalent temperature (PET) or the universal thermal climate index (UTCI). These are different physical quantities, measured differently, with different magnitudes and different policy meanings.
>
> Where a source reports surface temperature or a comfort index, that is stated explicitly, and the value is used only for the *relative ranking* of intervention families or to support a caveat. It is never converted into a °C air-temperature claim.

This discipline is not pedantry. Conflating the three metrics is the single most common way screening instruments mislead their users, and the divergence is large: Venter et al. (2021) report an approximately sixfold difference between surface and canopy-layer heat island magnitudes, and Zölch et al. (2016) report comfort improvements from trees (13 % PET reduction) that have no direct air-temperature equivalent. A decision-maker reading “3 °C of cooling” assumes air temperature. The methodology gives them air temperature, and where it cannot, it says so.

A direct consequence is that the tool systematically *understates* the benefit of shade-dominant interventions, because radiant load reduction — the mechanism through which shade most improves human thermal comfort — does not appear in an air-temperature figure. This is recorded as a limitation in Section 12.3 rather than corrected by an unsourced adjustment.

6.3 Adopted values

Table 3 gives the adopted values for all fourteen typologies. Derivations are summarised in Section 6.4 and reproduced in full, with citations, in Appendix A.

6.4 How envelopes were derived

Figure 4 shows the principal cross-typology anchor: the per-measure daytime central estimates reported by Keravec-Balbot et al. (2026), with two contrasting estimates overlaid.

Three principles governed the derivation of an envelope from these estimates.

An envelope is not a prediction. The lower bound represents a poorly performing instance of the typology; the upper bound represents a well-executed instance under favourable conditions. Site adjustment moves an assessment *within* this envelope (Section 7.6); it does not extend it.

Central estimates are ceilings, not midpoints. Because published central estimates already describe interventions that were built, studied, and in many cases modelled under favourable assumptions, adopted upper bounds sit at or below the synthesis central estimate rather than above it.

Where sources disagree, the conservative value is adopted and the disagreement is documented rather than averaged away. Two typologies — green roof and green façade — rest on

Table 3: The fourteen NbS typologies with their adopted values at methodology version 2026.08.02. All temperature envelopes are daytime, pedestrian-level *air* temperature reductions in degrees Celsius. The base cooling score is a 0–100 index expressing relative expected cooling performance before site adjustment; it is not a temperature. Evidence confidence is the rating defined in Section 8.2.

ID	Typology	Base score	Envelope (°C)	Confidence
Green				
01	Street tree planting	75	0.5 – 3.0	High
02	Shaded pedestrian corridor	80	0.5 – 3.0	Medium
03	Pocket park	65	0.3 – 1.5	High
04	Urban forest / dense canopy	90	1.0 – 3.0	High
05	Park upgrade	70	0.5 – 2.0	High
06	Schoolyard greening	75	0.5 – 2.0	Medium
Building				
07	Green roof	45	0.1 – 1.0	Medium
08	Green façade	50	0.3 – 2.0	Low
Blue-green				
09	Rain garden / bioswale	55	0.1 – 0.8	Low
10	Blue-green corridor	85	1.0 – 3.0	Medium
11	Riparian restoration	85	1.0 – 3.0	Medium
Hybrid				
12	Permeable shaded plaza	70	0.5 – 2.5	Medium
13	Courtyard greening	60	0.3 – 1.5	Low
14	Mixed NbS package	80	1.0 – 3.0	Medium

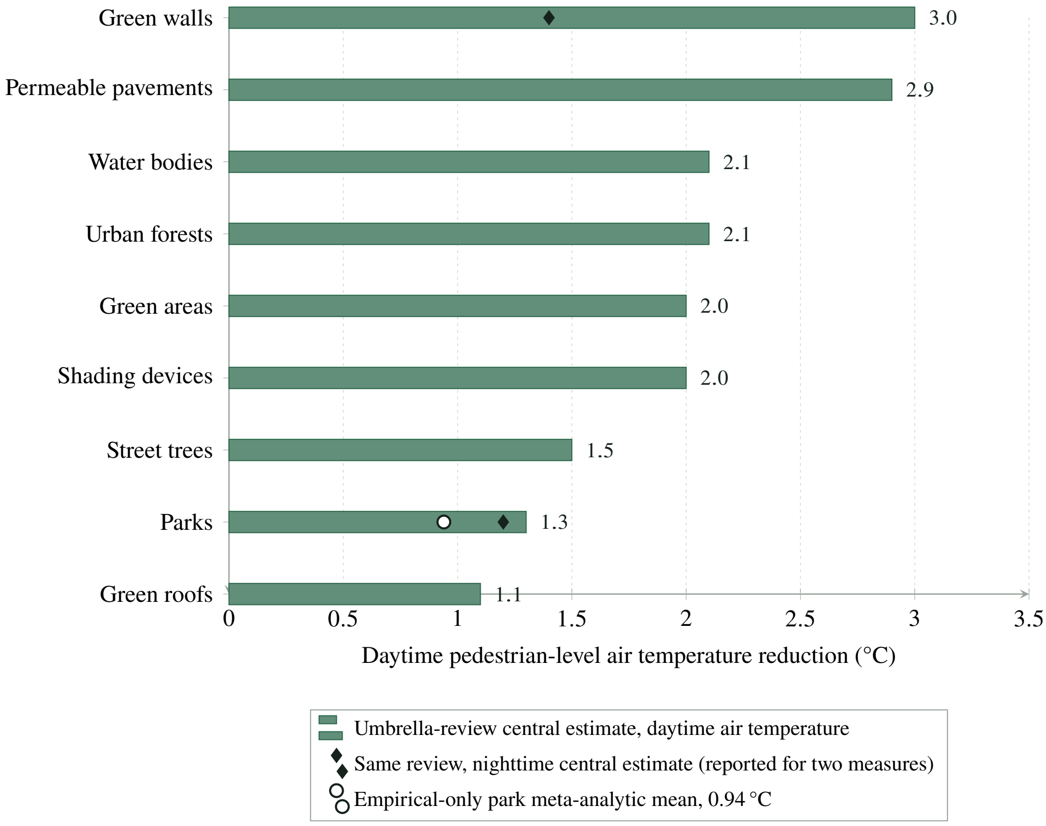

Figure 4: Central estimates of daytime pedestrian-level *air* temperature reduction by measure, from the umbrella review of 64 systematic reviews by Keravec-Balbot et al. (2026). Overlaid markers show, where an alternative estimate exists for the same measure, the nighttime central estimate from the same review and the empirical-only park meta-analytic mean of 0.94 °C reported by Bowler et al. (2010). The gap between the park bar (1.3 °C, modelling-inclusive) and the open marker (0.94 °C, empirical only) is the modelling bias discussed in Section 6.5.1; the diamonds show how much smaller nighttime effects are (Section 12.2).

genuinely conflicting evidence, and both are treated in Section 6.5.3.

6.5 Three calibration decisions that warrant reviewer attention

Three decisions materially lowered the tool's headline numbers relative to a naive reading of the literature. Each is stated here because each is contestable.

6.5.1 Modelling bias is corrected downward

Keravec-Balbot et al. (2026) synthesises 64 systematic reviews, many of them dominated by simulation studies, and simulation systematically reports larger cooling effects than field measurement. The magnitude of this bias is visible within the tool's own source set. For parks, the modelling-inclusive synthesis gives a central estimate of 1.3 °C, while Bowler et al. (2010), whose meta-analysis is restricted to empirical studies, gives 0.94 °C — roughly 72 % of the modelling-inclusive figure, for the one measure where both are available.

Adopted upper bounds therefore sit at or below synthesis central estimates rather than above them. The clearest case is street tree planting: the synthesis reports (4.0 ± 1.6) °C for street trees in tropical climates, but the adopted ceiling is 3.0 °C. The tropical figure carries a ±1.6 °C interval and describes favourable tropical conditions; adopting it as a general ceiling would overstate typical performance.

This correction is a judgment, not a calculation. A single ratio observed for a single measure is not a bias correction factor, and the methodology does not present it as one — it is used to justify the *direction* of the adjustment, not its size. A reviewer who considers the correction too aggressive, or not aggressive enough, is disputing exactly the right thing.

6.5.2 The mixed package does not claim super-additivity

An earlier draft of the methodology proposed a 3.5 °C ceiling for the mixed NbS package, on the intuition that combining measures should outperform any single measure. That value was rejected.

No retrieved source quantifies super-additive cooling from combining interventions. Gunawardena et al. (2017) describes green and blue features as offering synergistic benefits, but qualitatively and without quantification. Keravec-Balbot et al. (2026) reports no synthesis estimate for combined packages at all; its best-performing individual measures cluster at 2.0–3.0 °C.

The envelope is therefore capped at 1.0 ––3.0 °C, the best-evidenced single-measure ceiling. Claiming that a package outperforms its strongest component would be an unsourced assumption placed at exactly the point where a user is most likely to select it. The package's genuine advantage is expressed where evidence supports it — breadth of co-benefits and robustness across varying site conditions — and not as additional degrees.

6.5.3 Green roofs and green façades are the weakest points, and are labelled as such

Green roof. Three sources point in different directions. Keravec-Balbot et al. (2026) places green roofs at 1.1 °C, the lowest-performing green measure in the synthesis. Zölch et al. (2016) finds green roof effects at pedestrian level *negligible*. Santamouris (2014) reports a 0.3–3 K reduction in average urban ambient temperature — but the abstract states this explicitly for green roofs *applied on a city scale* in *simulation studies*.

That last qualification decides the matter. A site-level assessment evaluates one roof on one building, not mass deployment across a city, and the physics differ: a green roof's thermal benefit is concentrated at roof level and inside the building beneath, and it does not propagate to the street canyon in the way city-scale albedo and evapotranspiration changes would. The 0.3–3 K figure is therefore **not** used as a site-level typology value. The adopted envelope of 0.1–1.0 °C sits near the bottom of the published spread, and the tool states in its output that green roof benefits are principally building-level rather than street-level.

Green façade. Here the sources conflict outright. Keravec-Balbot et al. (2026) ranks green walls at the top of its air-temperature table with a 3.0 °C central estimate — but with an approximately fivefold spread across climate subzones (Figure 1). Zölch et al. (2016) ranks green façades *below* trees on thermal comfort, at 5–10 % PET reduction against 13 % for trees.

The two are not directly comparable, since one reports air temperature and the other a comfort index; but the ranking reversal is real and needs an explanation. The most plausible one is measurement position: near-wall measurements capture a strong local effect that does not extend across a street canyon. This is a hypothesis, not a finding from either source, and it is labelled as such.

Given an unresolved conflict between sources and a fivefold climate spread, the typology receives a conservative envelope of 0.3–2.0 °C, an evidence confidence rating of *low*, and an explicit caveat in the tool's output. Its confidence rating also caps the cooling-block confidence at medium regardless of how complete the user's inputs are (Section 8.3.1).

And one further weak point. Rain garden / bioswale deserves the same flag. No retrieved source quantifies air-temperature cooling for rain gardens or bioswales specifically. Keravec-Balbot et al. (2026) offers no bioswale-specific estimate, and its neighbouring categories — green areas at 2.0 °C, water bodies at 2.1 °C — are not transferable to a small drainage feature. Gunawardena et al. (2017) identifies tree-dominated greenspace as delivering heat-stress relief and does not identify small blue features as significant coolers. The adopted envelope of 0.1–0.8 °C is a reasoned lower bound from adjacent evidence, flagged low confidence, and the tool reports this typology's primary benefits as stormwater management and biodiversity rather than cooling.

6.6 Suitability conditions

Each typology declares conditions under which it is unsuitable: a minimum viable site area, a soil requirement, an irrigation requirement, and any unsuitable climate zones. Appendix A lists these per typology.

When a disqualifying condition is met, the tool computes the assessment transparently but attaches a prominent *"not suitable for this site"* flag naming the reason, which is carried into the recommendation text and the exported report. Silently returning a merely lower score for a physically implausible intervention — riparian restoration on a site with no water feature, for instance — would misrepresent the finding as a matter of degree when it is a matter of kind.

Unsuitable typologies nonetheless remain fully selectable. A user deliberately testing a hypothesis should not be overridden by the tool, and the flag follows the result through to the report regardless.

One suitability rule is worth singling out because it encodes a caveat that would otherwise be lost in prose. Gunawardena et al. (2017) notes that poorly designed bluespace may *exacerbate* heat stress under oppressive humid conditions. Rather than leaving this as a written warning, the methodology encodes it as a climate-suitability penalty: blue-green typologies are downgraded to the *moderate* condition level in tropical-wet zones (Table 6). A caveat that changes a number is a caveat that gets read.

7 Scoring formulas

7.1 Notation

All scores are on a 0–100 scale unless stated otherwise. Two bounding operators are used throughout:

$$\mathrm{clamp}(x) = \min\big(100,\ \max(0,\ x)\big), \tag{2}$$

$$\mathrm{clip}(x,\ l,\ u) = \min\big(u,\ \max(l,\ x)\big). \tag{3}$$

clamp bounds a score to the reporting scale; clip bounds a physical estimate to a stated envelope. The distinction matters in Section 7.6, where a score is clamped to [0, 100] while a temperature is clipped to a literature-derived interval.

Every weight appearing below is reproduced with its rationale in Appendix B. Within each weighted sum, the weights sum to unity.

7.2 Heat Exposure Score

There are two paths, selected by data availability.

Data-rich path. Where a land-surface-temperature anomaly is supplied and converted to S_{LST} by Equation (1):

$$S_{\mathrm{heat}} \ = \ \mathrm{clamp}\big(0.40\, S_{\mathrm{LST}} + 0.25\, S_{\mathrm{imp}} + 0.20\, S_{\mathrm{solar}} + 0.15\, S_{\mathrm{vegdef}}\big), \tag{4}$$

where S_{imp} is the impervious surface percentage of the site used directly as a 0–100 score, S_{solar} is the qualitative solar exposure level normalised by Table 2, and the vegetation deficit is

$$S_{\mathrm{vegdef}} \ = \ \mathrm{clamp}(100 - g)\,, \tag{5}$$

with g the existing green cover percentage.

Data-poor path. Where no LST anomaly is supplied, the Heat Exposure Score is the normalised qualitative heat exposure level directly: $S_{\text{heat}} = S_{\text{heatlevel}}$.

Basis. The component selection reflects established surface-energy-balance drivers of urban heat: surface temperature, sealed surfaces, solar loading, and vegetation deficit. Ziter et al. (2019) provides direct empirical support for two of the four — impervious cover raises air temperature approximately linearly, and canopy cover reduces it nonlinearly — and additionally establishes that the warming effect of impervious surfaces persists at night while canopy cooling largely does not.

What is expert judgment here. The *relative* weights are not derived from literature. No retrieved source ranks the marginal contribution of LST anomaly against imperviousness against solar loading against vegetation deficit for the purpose of a screening index. The weights express a judgment that a directly measured thermal signal should dominate where it is available, that sealed surface is the strongest single proxy where it is not, and that vegetation deficit — already partly captured by the other three — should carry the least. This judgment is declared, versioned, and subject to the sensitivity analysis of Section 9.

7.3 Vulnerability Score

$$S_{\text{vuln}} = \text{clamp}\big(0.40\, S_{\text{dens}} + 0.40\, S_{\text{groups}} + 0.20\, S_{\text{coolaccess}}\big), \tag{6}$$

where S_{dens} is normalised population density, S_{groups} is normalised vulnerable-group presence, and $S_{\text{coolaccess}}$ is the cooling access *deficit* obtained through the inverted mapping of Section 5.3.

Basis. Indicator selection follows Reid et al. (2009), whose factor analysis of ten heat-vulnerability variables identified four factors explaining more than 75 % of total variance: social and environmental vulnerability, social isolation, air-conditioning prevalence, and the proportion of elderly and diabetic residents.

Population density and vulnerable-group presence are weighted *equally* because that analysis provides no basis for ranking demographic vulnerability above exposure density; asserting a difference would be inventing a finding. Cooling access carries the smaller weight for two stated reasons: it is a single indicator where the other two are composites, and it is the field most frequently unknown at screening stage, so a larger weight would amplify the neutral default of Section 5.4.

7.4 Heat Priority Index

$$\text{HPI} = \text{clamp}\big(0.60\, S_{\text{heat}} + 0.40\, S_{\text{vuln}}\big). \tag{7}$$

The index is interpreted by the bands in Table 4. Band boundaries drive recommendation text, which is why their stability under weight perturbation is one of the four quantities the sensitivity analysis reports (Section 9.4).

Table 4: Interpretation bands for the Heat Priority Index and the NbS Cooling Opportunity Score.

Range	Heat Priority Index	Opportunity Score
0 – 30	Low	Low priority
31 – 60	Medium	Moderate opportunity
61 – 80	High	Strong opportunity
81 – 100	Critical	High-priority project

7.5 Derived site adjustment factor

Four site conditions modulate typology performance. Each condition resolves to a level, and each level maps to a multiplicative factor:

Condition level	Poor	Moderate	Good	Excellent	Unknown
Factor	0.5	0.8	1.0	1.2	1.0

The composite adjustment factor is

$$A = 0.40\, f_{\text{canopy}} + 0.25\, f_{\text{soilwater}} + 0.20\, f_{\text{scale}} + 0.15\, f_{\text{climate}}, \tag{8}$$

which ranges over $[0.5, 1.2]$, attaining its bounds only when all four conditions are poor or all four are excellent.

All four conditions are derived, not asked. No additional questions are put to the user; each condition is computed from inputs already supplied. Table 5 gives the rules.

Table 5: Derivation of the four site adjustment conditions from inputs the user has already supplied.

Condition	Derivation rule
Canopy	$c = \dfrac{\text{existing tree canopy + new canopy at maturity}}{\text{site area}} \times 100$, then: poor if $c < 10$; moderate if $10 \le c < 25$; good if $25 \le c < 40$; excellent if $c \ge 40$.
Soil–water	Soil availability and irrigation availability are each mapped to a condition level, and the *more limiting* of the two is taken. The mapping is *none* → poor; *limited*, *occasional* → moderate; *moderate* → good; *reliable*, *high* → excellent. When exactly one of the two is known, the condition is capped at *good*: a pair containing an unknown may never exceed the neutral factor, so declaring reliable irrigation is never worse — and with high soil strictly better — than skipping the question (D-026, version 2026.08.02).
Scale	From the declared assessment scale: *city*, *district* → excellent; *neighbourhood* → good; *site*, *building* → moderate.
Climate	A lookup of typology family against climate zone (Table 6).

Why canopy carries the largest weight. It is the condition with the strongest direct empirical support. Ziter et al. (2019) found that cooling increases nonlinearly with canopy cover and is greatest above approximately 40 % cover; the *excellent* boundary is placed at that empirically identified inflection rather than chosen for convenience. The same study locates maximum daytime cooling at 60–90 m radii, which is the basis for the scale condition: a district-scale or city-scale intervention is more likely to reach the spatial scale at which the effect is strongest than a single-building one.

Why climate is present at all. Keravec-Balbot et al. (2026) finds that adding Köppen–Geiger subzone raises explained variance in cooling effectiveness from 12 % to 78 %. Omitting climate would discard the largest single explanatory term in the available evidence.

Table 6: Climate suitability matrix: the condition level assigned to each typology family in each supported climate zone, resolved to a factor through the condition-level table above. Vegetation-based typologies are downgraded in arid and semi-arid zones where water limitation constrains evapotranspiration; blue-green typologies are downgraded in tropical-wet zones following the humidity caveat of Gunawardena et al. (2017), and upgraded in tropical-dry zones. An unrecognised zone yields *unknown*, whose factor is the neutral 1.0.

Climate zone	Green	Building	Blue-green	Hybrid
Tropical wet	Good	Good	Moderate	Good
Tropical dry	Good	Good	Excellent	Good
Arid	Moderate	Moderate	Moderate	Good
Semi-arid	Moderate	Good	Good	Good
Temperate	Good	Good	Good	Good
Other	Unknown	Unknown	Unknown	Unknown

7.6 Cooling Potential Score and temperature reduction

The Cooling Potential Score scales the typology's base cooling score by the site adjustment factor:

$$S_{\text{cool}} = \text{clamp}(B \times A)\,, \tag{9}$$

where B is the base cooling score from Table 3.

The reported temperature reduction range is computed differently, and the difference is the point of this subsection:

$$\Delta T_{\min} = \text{clip}(\tau_{\min} \times A,\ \tau_{\min},\ \tau_{\max})\,, \tag{10}$$

$$\Delta T_{\max} = \text{clip}(\tau_{\max} \times A,\ \tau_{\min},\ \tau_{\max})\,, \tag{11}$$

where $[\tau_{\min}, \tau_{\max}]$ is the typology's literature envelope from Table 3.

The clipping rule is a deliberate correction to the draft methodology. Without it, an adjustment factor above 1.0 applied to a typology's upper bound produces temperature claims exceeding anything in the retrieved literature. An urban forest on an excellent site ($A = 1.2$, $\tau_{\max} = 3.0$ °C) would report up to 3.6 °C — a figure no retrieved source supports for a site-level intervention.

Because the published envelopes already represent well-executed interventions under favourable conditions, favourable site conditions must not be permitted to compound that favourability. The 0–100 Cooling Potential Score of Equation (9) remains free to rise above the base value, so the tool can still express that a site is exceptionally well suited — but it expresses this as a higher relative score, not as a larger physical claim.

Figure 5 shows the behaviour across the full range of the adjustment factor. The rule is asymmetric in effect, and it is worth being precise about what it does. For $A < 1$ the upper bound falls — a poor site is reported as delivering less — while the lower bound is held at $\tau_{\min}$, so the reported range narrows from the top. For $A > 1$ the upper bound is held at $\tau_{\max}$ while the lower bound rises, so the range narrows from the bottom. In both directions the reported interval remains a subinterval of the literature envelope: site conditions shift an estimate *within* the published evidence, and never outside it.

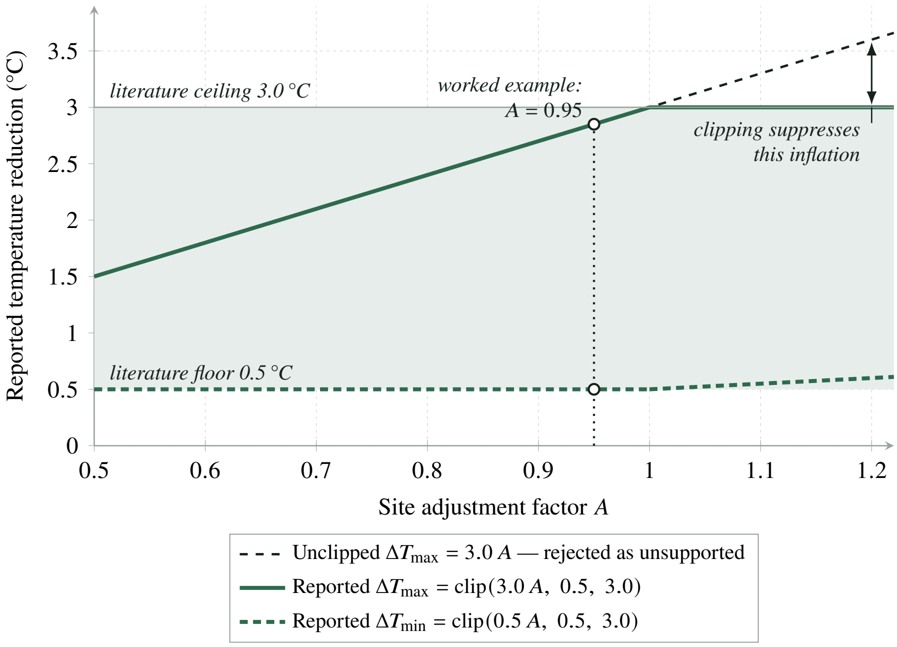

Figure 5: The temperature-envelope clipping rule of Equations (10) and (11), illustrated for street tree planting (envelope 0.5–3.0 °C) across the full 0.5–1.2 range of the site adjustment factor A. The dashed line is the unclipped product $3.0\,A$, which the methodology rejects because for $A > 1$ it exceeds the literature ceiling. The reported bounds remain within the shaded envelope throughout. The marked point locates the worked example of Section 11.

Derived reporting quantities. Two further quantities are reported directly. Shade potential is the fraction of the site brought under new canopy at maturity:

$$S_{\text{shade}} = \operatorname{clamp}\left(\frac{a_{\text{canopy}}}{a_{\text{site}}} \times 100\right), \tag{12}$$

with a_{canopy} the new canopy area at maturity and a_{site} the site area, both in square metres. And the midpoint of the adjusted temperature range, $\overline{\Delta T} = (\Delta T_{\min} + \Delta T_{\max})/2$, is assigned a heat-index improvement bucket: *low* below 0.5 °C, *medium* from 0.5 °C to 1.5 °C, and *high* from 1.5 °C upward.

Time to benefit. No maturity discount is applied to cooling values. Instead the expected maturity period is reported as a time-to-benefit class, so that a slow-maturing intervention is never presented as delivering its cooling immediately: under one year *immediate*, one to three years *short-term*, three to ten years *medium-term*, ten years and beyond *long-term* (half-open brackets). Absent a maturity period the output is *not estimated*.

7.7 NbS Suitability, co-benefit, and equity scores

Three further composite sub-scores enter the result. Their aggregation weights are declared in configuration and reproduced in Appendix B:

$$S_{\text{suit}} = \text{clamp}\big(0.30\,S_{\text{space}} + 0.25\,S_{\text{soil}} + 0.20\,S_{\text{water}} + 0.15\,S_{\text{maint}} + 0.10\,S_{\text{context}}\big), \tag{13}$$

$$S_{\text{cobenefit}} = \text{clamp}\big(0.25\,S_{\text{biodiv}} + 0.25\,S_{\text{storm}} + 0.20\,S_{\text{health}} + 0.15\,S_{\text{social}} + 0.15\,S_{\text{urbanq}}\big), \tag{14}$$

$$S_{\text{equity}} = \text{clamp}\big(0.35\,S_{\text{vulnben}} + 0.25\,S_{\text{access}} + 0.20\,S_{\text{safety}} + 0.20\,S_{\text{particip}}\big). \tag{15}$$

The co-benefit sub-indicators take per-typology default levels, each cited in the typology library (Appendix A), which the user may override.

The sub-indicator rules, fixed at version 2026.08.02. An earlier version of this document disclosed that the rules mapping user inputs to the individual sub-indicator scores were unspecified and would be fixed alongside the engine implementation. They are now fixed, live in `config/derived_scores.yaml`, and — like the aggregation weights — are declared expert judgment, versioned and adjustable rather than derived from literature. All suitability sub-indicators derive from inputs the user has already supplied; no new questions are asked.

Table 7: Derivation of the NbS Suitability sub-indicators (decision D-022). A score of 25 on space, soil, or water coincides with the hard "not suitable" flag of Section 6.6; an unknown availability never raises a flag, because a disqualification is never asserted from absent information.

Sub-indicator	Rule
Space	r = site area/typology minimum viable area: $r < 1 \rightarrow 25$ *and the flag*; $1 \leq r < 2 \rightarrow 50$; $2 \leq r < 5 \rightarrow 75$; $r \geq 5 \rightarrow 100$.
Soil	Availability against the typology requirement on the ordinal scale *none* < *limited* < *moderate* < *high*: requirement *none* → 100; availability unknown → 50; below requirement → 25 *and the flag*; meets exactly → 75; exceeds → 100.
Water	The same rule on *none* < *occasional* < *reliable* against the irrigation requirement.
Maintenance	Declared maintenance intensity through the inverted scale: low → 100, medium → 50, high → 25, unknown → 50.
Urban context	Site land use inside the typology's typical-use contexts → 100; outside them → 25 (a caution, not a disqualification); unknown → 50.

The four equity sub-indicators read how much equity benefit the intervention can deliver at this site, so a *deficit* raises the score, paralleling the risk framing of the Vulnerability Score: vulnerable-user benefit reuses the vulnerable-population input already collected; public accessibility and community participation are the two new optional inputs introduced at this version; and safety concern deliberately uses the standard (non-inverted) scale, because a site with greater safety and comfort problems stands to gain more from a well-designed intervention.

Two disclosures carry over unchanged from the previous version. The co-benefit weights include a *social inclusion* term for which the typology library supplies no default; absent a user override it falls to the neutral default of Section 5.4 and is itemised as an applied assumption. And the Equity Score of Equation (15) is computed and reported as an output block with its own confidence rating but does *not* enter the final aggregation of Section 7.11: equity influences the headline score through the Vulnerability Score instead. Whether that is the right design remains a legitimate question for reviewers.

7.8 Energy savings — derived, not assumed

An earlier draft of the methodology stored per-typology "energy reduction factors" in the range 2–15 %. No source could be found for any of them. They were removed and replaced by a derivation from the tool's own estimated cooling effect:

$$E = D \times \Delta T \times s, \qquad s \in [0.02,\ 0.04] \text{ per °C}, \tag{16}$$

where D is the user-supplied annual cooling energy demand in kW h and s is the temperature–electricity-demand sensitivity. Because both ΔT and s are intervals, the result is an interval:

$$E_{\min} = D \times \Delta T_{\min} \times 0.02, \qquad E_{\max} = D \times \Delta T_{\max} \times 0.04. \tag{17}$$

Basis. Akbari et al. (2001) reports that urban electricity demand increases by 2–4 % for each 1 °C of temperature increase, and estimates that 5–10 % of urban electricity demand serves to compensate for the 0.5–3.0 °C urban temperature elevation. Applying that sensitivity in reverse to an estimated temperature reduction yields an energy saving that is sourced, internally consistent with the tool's own cooling estimate, and automatically responsive to site conditions — a poorly suited site yields a smaller ΔT and therefore a smaller energy saving, with no separate parameter to maintain.

Preconditions. Energy savings are calculated only when all three of the following hold: the user confirms that nearby building cooling demand is relevant to the intervention; an annual cooling energy demand is supplied; and the selected typology is flagged as capable of affecting adjacent building loads. Otherwise the result carries an explicit status, checked in a fixed order — `typology_not_applicable` (the intervention cannot affect building loads, whatever the user supplied), `not_applicable` (the user states demand is not relevant), `relevance_not_confirmed` (the relevance question was skipped or answered *unknown*; a supplied demand figure cannot substitute for the unanswered confirmation), or `missing_energy_demand` — and **never a zero**, which would understate the result while appearing to be a finding.

An acknowledged weakness. The sensitivity is drawn from predominantly North American evidence, and its transferability to other building stocks, construction practices, and climates is untested. Building stock differs in envelope performance, in the prevalence and type of mechanical cooling, and in occupant behaviour, all of which affect how a change in outdoor air temperature translates into a change in electricity demand. This is an area where reviewer input is specifically sought. It is also worth noting that Akbari et al. (2001) is recorded in Appendix E as *metadata + finding (secondary)*: the bibliographic record and the 2–4 % figure were confirmed from multiple indexing records and an institutional listing, not from a retrieved full text.

7.9 Greenhouse gas emissions avoided

$$G = E \times \phi, \tag{18}$$

where ϕ is the grid emission factor in kg h/kW of carbon dioxide equivalent, applied to both ends of the energy interval.

Grid emission factors are sourced per country from Ember (2026), which publishes electricity carbon intensity for 215 countries under a CC-BY-4.0 licence permitting redistribution with attribution inside an Apache-2.0 tool. Where no verified factor is available for a country, the tool reports greenhouse gas savings as *not calculated* rather than substituting a global average.

At version 2026.08.02 no country emission factors ship with the tool. The country defaults file declares the source, licence, unit, and per-entry schema, but its country table is empty pending per-country verification. In consequence, a deployment using the shipped configuration unmodified will report greenhouse gas savings as *not calculated* for every country. This is the intended failure mode of the verification policy — no unverified value ships — but it means

Equation (18) is at present a specified formula awaiting its data, and any assessment reporting a greenhouse gas figure is necessarily using a locally supplied factor.

7.10 Costs — why the tool ships no default values

World Bank (2021) states that nature-based solution project costs vary significantly and are highly site- and project-specific, that unit costs differ sharply between developed and developing contexts — illustrated by dredging at 2 US$/m^3 in Bangladesh against 59 US$/m^3 in the United Kingdom, roughly a thirtyfold difference for the same operation — and that urban implementation costs typically exceed rural ones. No source was found offering globally applicable unit costs for the typologies in this library.

The tool therefore ships **no default cost values** in version 1. Where a user supplies a capital cost, the tool computes cost outputs; where they do not, capital cost, payback, and cost feasibility are all reported as *not estimated*.

The reasoning is worth stating explicitly, because shipping a plausible default would have been easy and would have made the tool look more complete. Fabricating a default per-square-metre cost would generate the most decision-relevant number in the entire assessment — simple payback — from an invented input. No confidence rating adequately qualifies a payback figure whose numerator was guessed: the reader sees a number of years, and the number of years is what enters the decision. Locally calibrated cost tables remain a defined extension point; a deployment may supply its own cost configuration, which the tool records as a documented assumption in the result.

Where costs are supplied:

$$C_{\text{annual}} = E \times p, \tag{19}$$

$$T_{\text{payback}} = \frac{C_{\text{capital}}}{C_{\text{annual}}}, \qquad \text{undefined if } C_{\text{annual}} \le 0, \tag{20}$$

with p the energy price per kW h. Because E is an interval, so is the payback, with the bounds inverted: the largest energy saving gives the shortest payback.

Cost feasibility is *derived* rather than user-asserted:

$$S_{\text{cost}} = f\big(\text{payback bracket},\ \text{implementation complexity},\ \text{maintenance intensity}\big), \tag{21}$$

with short payback, low complexity, and low maintenance yielding high feasibility. The bracket boundaries and combination rule, fixed at version 2026.08.02 (decision D-023), are as follows. Because the energy saving is an interval spanning an order of magnitude, the payback bracket is applied to the payback computed from the *central* energy estimate — the midpoint of the savings range — while the reported payback interval still carries both ends. The brackets are half-open and follow common public-investment screening horizons: under 5 years scores 100 (*short*); 5–10 years 75 (*medium*); 10–20 years 50 (*long*); 20 years and beyond 25 (*very long*). The combination is

$$S_{\text{cost}} = 0.50\, S_{\text{bracket}} + 0.25\, S_{\text{complexity}} + 0.25\, S_{\text{maintenance}}, \tag{22}$$

with complexity and maintenance normalised through the inverted scale so that low complexity and low maintenance raise feasibility, and unknowns taking the neutral 50, itemised. Payback carries half the weight as the only quantitative economic signal. An *investment readiness* level is reported alongside: the bracket sets the base (*short* → high, *medium* → medium, otherwise low), downgraded one level for high implementation complexity and one level for low energy-block confidence, with a floor at low; it is *not estimated* whenever cost feasibility is.

Exclusion rather than substitution. When payback cannot be computed, cost feasibility is reported as unavailable and is *excluded* from the final aggregation, with the remaining weights redistributed proportionally (Equation (24)). The alternative — substituting a neutral 50 — would silently reward projects that have provided no economic evidence at all, placing them ahead of projects that supplied a genuinely poor payback figure. Exclusion preserves the ordering among options that did supply data, and the missing block is reported rather than hidden.

7.11 Aggregation and the weighting question

The headline score is a weighted sum of six components:

$$S_{\text{opp}} = 0.25\,\text{HPI} + 0.25\, S_{\text{cool}} + 0.15\, S_{\text{suit}} + 0.15\, S_{\text{vuln}} + 0.10\, S_{\text{cobenefit}} + 0.10\, S_{\text{cost}}. \tag{23}$$

Where a component is unavailable — in practice, cost feasibility — the sum is taken over the available set $\mathcal{A}$ and renormalised:

$$S_{\text{opp}} = \frac{\sum_{i \in \mathcal{A}} w_i\, S_i}{\sum_{i \in \mathcal{A}} w_i}. \tag{24}$$

The result is interpreted through the bands of Table 4.

These weights are expert judgment, and the tool says so. Nardo et al. (2008) is explicit that weighting in composite indicators is a value choice rather than a technical derivation, and no literature can supply the relative importance of cooling performance against equity against cost. The weights are declared here, versioned in configuration, adjustable by any deployment, and are to be defended empirically through the sensitivity analysis specified in Section 9 rather than by appeal to authority.

7.12 The equity-forward weighting, stated openly

Vulnerability enters the final score *twice*. It contributes 0.40 of the Heat Priority Index, which itself carries 0.25 in Equation (23), and it appears again directly at 0.15. Decomposing the Heat Priority Index in Equation (23) gives

$$
\begin{aligned}
S_{\text{opp}} &= 0.25\big(0.60\,S_{\text{heat}} + 0.40\,S_{\text{vuln}}\big) + 0.25\,S_{\text{cool}} + 0.15\,S_{\text{suit}} + 0.15\,S_{\text{vuln}} + 0.10\,S_{\text{cobenefit}} + 0.10\,S_{\text{cost}} \\
&= 0.15\,S_{\text{heat}} + 0.25\,S_{\text{vuln}} + 0.25\,S_{\text{cool}} + 0.15\,S_{\text{suit}} + 0.10\,S_{\text{cobenefit}} + 0.10\,S_{\text{cost}}. \qquad (25)
\end{aligned}
$$

The effective weights are set out in Table 8.

Table 8: Effective weights in the NbS Cooling Opportunity Score, after decomposing the Heat Priority Index. The nominal weights are those of Equation (23); the effective weights are those of Equation (25). Both columns sum to unity, as the total row records. The two rows below that total are not additional weights: they restate, for emphasis, the totals that vulnerability and heat exposure reach once the Heat Priority Index is decomposed.

Component	Nominal weight	Effective weight
Heat Priority Index	0.25	—
of which heat exposure		0.15
of which vulnerability		0.10
Cooling Potential	0.25	0.25
NbS Suitability	0.15	0.15
Vulnerability (direct)	0.15	0.15
Co-benefits	0.10	0.10
Cost Feasibility	0.10	0.10
Total	**1.00**	**1.00**
Restated from the rows above, not added to them:		
Vulnerability, total		**0.25**
Heat exposure, total		**0.15**

The tool therefore weights *who is affected* (0.25) above *how hot the site is physically* (0.15). Between two equally hot sites, the one serving a more vulnerable population ranks higher.

This is a value position, not a technical necessity. There is no finding in the literature that requires it, and a reasonable reviewer may reject it. A city whose mandate is to reduce peak thermal load across its territory, rather than to direct adaptation investment toward disadvantaged communities, would be right to rebalance it.

It is disclosed here, and quantified, for a specific reason: an undisclosed double contribution would be a defect, indistinguishable from an arithmetic error, and a reviewer discovering it unaided would be entitled to distrust everything else in the methodology. Disclosed and quantified, it is a defensible position that reviewers may legitimately dispute. Any deployment may rebalance it in `config/weights.yaml` without touching code, and every result is to record the methodology version under which it was produced.

The double contribution is also not an accident of construction that happened to be noticed afterwards. It follows from a deliberate decision, recorded in the project's decision log as D-007, that the tool should prioritise who is affected slightly above how hot a place is. The alternative designs — dropping the direct vulnerability term, or removing vulnerability from the Heat Priority Index — were both available; each would have produced a defensible instrument with a different value stance.

8 Uncertainty, confidence, and missing data

8.1 Ranges, not point estimates

All quantitative outputs — temperature reduction, energy savings, greenhouse gases avoided, cost savings, payback — are reported as ranges. This is not a presentational preference; it is what the evidence supports.

The adopted cooling envelopes of Table 3 span factors of three to ten: the narrowest are threefold (urban forest, blue-green corridor, riparian restoration, and the mixed package, each 1.0–3.0 °C), while green roof at 0.1–1.0 °C is tenfold and rain garden / bioswale at 0.1–0.8 °C is eightfold. The climate-differentiated estimates in Keravec-Balbot et al. (2026) carry their own uncertainty on top of that: street trees in arid climates are reported at (1.7 ± 1.4) °C, an interval nearly as wide as the estimate itself. A single figure drawn from within a spread of that magnitude would imply a precision that does not exist, and would be quoted, in practice, without its provenance.

The ranges compound. An energy estimate combines a temperature interval with a sensitivity interval (Equation (17)), so its width is the product of the two — the worked example in Section 11 produces an energy range spanning roughly an order of magnitude. That width is uncomfortable to present and it is the honest answer. A user who needs a tighter energy figure needs a building energy model, which is precisely the boundary Section 2.2 draws.

8.2 Confidence levels

Two distinct things are called confidence in this methodology, and separating them prevents a common confusion.

Evidence confidence is a property of a *typology*, recorded in the library, expressing how well grounded its adopted values are:

High multiple converging sources, including a synthesis.

Medium a synthesis estimate with limited typology-specific corroboration.

Low sparse, indirect, or conflicting evidence. The tool flags these results explicitly in its output.

Block confidence is a property of an *assessment*, expressing how completely the user filled in the inputs that a given output block depends on. It is computed as described below.

A typology may have high evidence confidence in an assessment with low block confidence, or the reverse. Reporting a single merged figure would lose the distinction between "we know this intervention well but you told us little about your site" and "you described your site thoroughly but the evidence for this intervention is thin". Those call for different responses.

8.3 Branched confidence

A single confidence rating would be dominated by whichever input block happened to be least complete, and would obscure the common case of a site with good physical data and no economic data. Confidence is therefore computed separately for each output block — cooling, energy, economic, and equity — from the completeness of the inputs feeding that block:

Table 9: The block confidence model. Completeness is the proportion of the fields relevant to that output block which the user supplied.

Completeness of relevant fields	Confidence	How the result should be read
Below 40 %	Low	A screening signal for further investigation
40–70 %	Medium	Indicative; the main drivers are known
Above 70 %	High	As reliable as a screening instrument gets

The fields counted for each block are enumerated in `config/derived_scores.yaml` (fixed at version 2026.08.02, decision D-024): ten slots for cooling — with the LST anomaly and the qualitative heat exposure level sharing a single either-of slot, since they are alternative signals for the same quantity — three for energy, four for economic, and six for equity. A field answered *unknown* counts as **not supplied**: an explicit unknown carries no more information than a skipped question. The thresholds apply with exact boundaries (below 40 % low, 40–70 % inclusive medium, above 70 % high), and the overall rating attached to the headline score is the *lower median* of the four block ratings — exact, slightly conservative, and never higher than three of the four blocks.

Two additional constraints apply.

8.3.1 Evidence confidence propagates

A typology rated *low* evidence confidence in the library — green façade, rain garden / bioswale, and courtyard greening at version 2026.08.02 — caps the cooling-block confidence at *medium* regardless of how complete the user's inputs are.

The reason is straightforward: complete inputs cannot compensate for thin underlying evidence. A user who answers every question about a site where a green façade is proposed has told the tool a great deal about the site and nothing at all about whether green façades reliably reduce street-level air temperature, which is the question the disagreement between Keravec-Balbot et al. (2026) and Zölch et al. (2016) leaves open. The interface states the reason for the cap explicitly rather than showing an unexplained ceiling.

8.3.2 A high score with low confidence is a flag, not a verdict

The combination of a high Opportunity Score and low block confidence is presented as an invitation to investigate further, never as a conclusion. In practice this is the most dangerous output the tool can produce, because the headline number is exactly what gets extracted and quoted. The interface and the exported report attach the confidence rating to the score wherever the score appears, and Section 12.9 identifies quoting a score without its confidence as a misuse.

8.4 Missing data

Required inputs must be present; the assessment does not proceed without them. Optional inputs, when absent, follow explicit rules:

- **Qualitative inputs** default to *unknown*, which normalises to the neutral 50 of Section 5.4.
- **Quantitative inputs** propagate as *not calculated*. They are **never** treated as zero. A zero would understate the result while appearing to be a finding — an energy saving reported as 0 kW h reads as "this intervention saves no energy", which is a claim, whereas the truth is "we were not told the demand", which is an absence.
- **Derived outputs** whose inputs are missing carry an explicit status naming the reason, as the energy statuses of Section 7.8 illustrate.

Every default the engine applies is itemised in the result and reproduced in the exported report, so that a reader can see exactly which numbers came from the user and which from the tool. This list is not a diagnostic aid; it is part of the output, and a report circulated without it has removed the information needed to judge the numbers it contains.

9 Sensitivity analysis

This section specifies the design and reports its results.

The design below was published at methodology version 2026.07.30, before the calculation engine existed, precisely so that it could be criticised before execution and so that the results could not be shaped after the fact to suit the outcome. The analysis was executed in Phase 2 against the twenty hand-verified golden scenarios; Section 9.4 reports the results at version 2026.08.02, produced by `tools/sensitivity_analysis.py` and regenerated whenever any aggregation weight changes.

9.1 Why sensitivity analysis is obligatory here

Nardo et al. (2008) treats uncertainty and sensitivity analysis as integral to a credible composite indicator rather than as an optional addition. The argument applies with particular force to this methodology.

The aggregation weights of Equation (23) are the tool's most consequential unsourced choice. Every other quantitative element — the typology envelopes, the canopy threshold, the energy sensitivity, the LST normalisation — traces to a published source that a reviewer can consult and dispute on its own terms. The weights trace to a judgment. That judgment determines the headline number, and therefore determines which projects a city funds.

Two responses to this are available. One is to argue that the weights are reasonable and hope the reader agrees. The other is to measure how much the output actually depends on them and publish the answer, including if the answer is inconvenient. Only the second is checkable, and it is the one adopted.

There is a further reason specific to this tool. If rank ordering turns out to be robust to weight perturbation, then the disclosed equity-forward stance of Section 7.12 — while remaining a genuine value choice — changes fewer decisions than its prominence suggests, and deployments can adopt the tool without first having to resolve a philosophical dispute. If ordering turns out to be fragile, users need to know that a different but equally defensible weight set would have produced a different priority list, and the tool should say so at the point of use. Both outcomes are informative; only running the analysis distinguishes them.

9.2 Design of the analysis

Each aggregation weight in Equation (23) is varied by ± 25 % of its nominal value, with the remaining weights renormalised to preserve a unit sum, and the full golden-scenario set is re-scored under each perturbation. Formally, for the perturbation of weight w_k by a factor $\lambda \in \{0.75,\ 1.25\}$:

$$w'_k = \frac{\lambda w_k}{\lambda w_k + \sum_{i \neq k} w_i}, \qquad w'_j = \frac{w_j}{\lambda w_k + \sum_{i \neq k} w_i} \text{ for } j \neq k. \tag{26}$$

The golden-scenario set is the same collection of hand-verified input/output cases that serves as the engine's regression suite (Section 10.5), so the scenarios exercised by the sensitivity analysis are the same ones whose correctness is independently asserted.

9.3 What is reported

Four quantities are reported for each perturbation, chosen because they correspond to four distinct ways the tool could mislead a user.

1. **Rank stability** — the proportion of scenario pairs whose relative ordering is preserved under the perturbation. This is the primary measure. Prioritisation is the tool's actual function, so ordering matters more than absolute values: a perturbation that shifts every score by five points but reorders nothing has not changed a single decision.
2. **Score displacement** — the distribution of the absolute change in the Opportunity Score across the scenario set, reported as a distribution rather than a mean, because a small average displacement concealing a few large ones is a materially different result from a uniformly small one.
3. **Category migration** — how often a scenario crosses a band boundary in Table 4, for instance from *Strong opportunity* to *Moderate opportunity*. Categories drive the recommendation text, so a category change is a change in what the tool *says*, not merely in what it scores. A score displacement of two points matters greatly at 80.5 and not at all at 45.
4. **Influence ranking** — which weights the output is most sensitive to, ordered. This tells a deployment considering recalibration which weight to argue about first, and it identifies which of the tool's value judgments are actually load-bearing.

9.4 Results at version 2026.08.02

The full scenario set (twenty scenarios, 190 scenario pairs) was re-scored under each of the twelve perturbations — six weights, each at $\lambda \in \{0.75, 1.25\}$ of Equation (26).

Table 10: Sensitivity of the NbS Cooling Opportunity Score to ±25 % perturbation of each aggregation weight, across the twenty golden scenarios. Rank stability is the proportion of the 190 scenario pairs whose ordering is preserved; displacement is the absolute change in the Opportunity Score (0–100 scale); migrations counts scenarios whose interpretation band changed.

Weight	Shift	Rank stab.	Mean disp.	Max disp.	Migrations
Heat Priority Index	−25%	0.9737	0.56	2.00	1
	+25%	0.9895	0.49	1.77	0
Cooling Potential	−25%	0.9895	0.62	2.01	1
	+25%	0.9842	0.54	1.77	1
NbS Suitability	−25%	0.9895	0.47	1.49	0
	+25%	0.9842	0.43	1.38	0
Vulnerability	−25%	0.9895	0.52	1.75	0
	+25%	0.9895	0.47	1.62	0
Co-benefits	−25%	0.9947	0.32	0.77	0
	+25%	0.9895	0.30	0.74	0
Cost Feasibility	−25%	0.9895	0.17	1.12	0
	+25%	0.9842	0.16	1.06	0

The four committed quantities, in the order of Section 9.3:

1. **Rank stability.** Worst case 0.973 7, under the −25% Heat Priority Index perturbation: five of 190 pairs reorder. Nine of the twelve perturbations preserve at least 98.4 % of pair orderings. Every reordering involves a pair whose baseline scores differ by less than about two points — pairs the methodology itself would describe as materially equivalent.
2. **Score displacement.** Pooled over all 240 scenario-perturbation combinations: mean 0.42 points, median 0.32, 75th percentile 0.59, maximum 2.01 on the 0–100 scale.
3. **Category migration.** Three of 240 combinations cross a band boundary, each from a baseline score within about 1.3 points of the boundary (59.58 → 60.3 and 60.39 across the Moderate–Strong line; 80.92 → 79.63 across the Strong–High-priority line). No scenario moves by more than one band, and no scenario distant from a boundary migrates.
4. **Influence ranking.** By mean absolute displacement: Cooling Potential (0.58), Heat Priority Index (0.53), Vulnerability (0.49), NbS Suitability (0.45), Co-benefits (0.31), Cost Feasibility (0.17). Cost feasibility ranks last partly by construction: it is excluded and its weight redistributed in the fifteen of twenty scenarios that supply no cost data, so the weight binds only where economic evidence exists.

Interpretation. Under ±25 % perturbation of any single aggregation weight, the tool's priority ordering is substantially stable: at least 97.4 % of pairwise orderings survive every perturbation, scores move by well under half a point on average, and category changes occur only for sites already sitting on a band boundary. The disclosed equity-forward stance of Section 7.12 therefore changes fewer decisions than its prominence suggests: a deployment that rebalances the weights within this range should expect a materially similar priority list unless two options were near-tied to begin with — and near-ties are exactly the cases the methodology already tells users not to resolve on the headline score alone.

Two qualifications are owed. The analysis perturbs one weight at a time, so these figures are not bounds for simultaneous re-weightings. And the golden scenarios are designed to span the methodology's behaviour — climates, typologies, data completeness, cost availability — not to be a probability sample of real assessments; the stability figures are properties of this set. Both qualifications argue for reading near-boundary results as ranges rather than verdicts, which the confidence mechanism already enforces at the point of use.

9.5 Publication and maintenance

The results above are also committed to the repository in machine-generated form (`docs/methodology/SENSITIVITY-` and in the Methodology Report, and must be regenerated whenever any weight changes. Because a weight change requires a methodology version bump (Section 13), and because the version stamp is enforced across all configuration files by continuous integration, a weight change that does not carry a regenerated sensitivity analysis is visible in review rather than silent.

The threshold at which a result would prompt a methodology change is stated in advance to avoid its being negotiated afterwards: if rank stability under any single ± 25 % weight perturbation falls below a level at which the tool's prioritisation could be considered reliable, the appropriate response is to report that limitation prominently at the point of use — telling the user that their priority ordering is weight-sensitive — rather than to adjust the weights until the number improves.

10 Implementation and reproducibility

This section describes how the methodology is realised in software. It is not an appendix to the science: it is the part of the paper that makes the preceding claims *checkable* rather than merely asserted. A methodology whose values live in prose can drift from the code that applies them, and a reader has no way to detect the drift. The arrangement described here is designed so that such drift fails the build.

10.1 Configuration as data

Every methodology value lives in schema-validated, version-stamped YAML under `config/`:

Table 11: The methodology configuration files.

File	Content
`nbs_typologies.yaml`	The fourteen typologies: base cooling scores, temperature envelopes, evidence confidence, cooling mechanisms, suitability conditions, co-benefit defaults, and a `sources` array carrying, for each citation, the source key and the specific finding it supports.
`weights.yaml`	Every weighted sum in Section 7, with its rationale, plus the sensitivity-analysis perturbation parameter.
`adjustment_factors.yaml`	The condition-level to factor table, the four derivation rules of Table 5, and the climate suitability matrix of Table 6.
`input_mapping.yaml`	Qualitative to numeric normalisation, the inverted scale, the LST normalisation, and the heat-index improvement buckets.
`energy_model.yaml`	The cooling-energy derivation of Section 7.8: the formula, the sensitivity interval of 0.02–0.04 per °C with its citation, the three preconditions, the reported statuses, and the recorded limitation on transferability.
`country_defaults.yaml`	Grid emission factor source, licence, and per-country schema; energy price policy. Deliberately contains no cost values.
`recommendation_templates.yaml`	The deterministic text fragments from which recommendations are composed, and the conditions under which climate-dependent caveats fire.
`derived_scores.yaml`	The derivation rules fixed at version 2026.08.02: suitability and equity sub-indicator rules, cost-feasibility payback brackets, investment-readiness rules, time-to-benefit buckets, the confidence field-to-block mapping, and the score interpretation bands.

The scoring code contains no typology value, no weight, and no threshold. Changing the methodology never requires changing code, which has three consequences. A deployment can rebalance the weights it disagrees with — including the equity-forward stance of Section 7.12 — without forking the software. A reviewer can read the complete parameterisation of the tool in seven short files rather than by tracing constants through source code. And the diff of a methodology change is legible: it shows values moving, not logic moving.

10.2 What is implemented at version 2026.08.02

This section distinguishes throughout between what the repository *contains* at methodology version 2026.08.02 and what it *specifies* for the next development phase. The distinction is stated because a reviewer can and should check it by cloning the repository, and because a paper that described planned software in the present tense would forfeit precisely the credibility this section exists to establish.

Implemented and running in continuous integration: the configuration layer — schema-validated loading of all eight methodology files, version lock-step enforcement, and the citation cross-check against the bibliography — the six evidence rules of Section 10.4; and, as of version 2026.08.02, the calculation engine itself: the input and result schemas, one scoring module per formula family, branched confidence, the deterministic recommendation composer, and the orchestrating entry point, under a 100 % line-coverage gate with twenty hand-verified golden scenarios, a byte-identical determinism test, and the executed sensitivity analysis of Section 9.4.

Specified but not yet implemented: the API layer and the web interface. These are Phases 3 and 4.

Every number in the worked example of Section 11 was first computed by hand from the specification and is now also asserted, digit for digit, by the engine's golden-scenario suite — the hand derivation is the expected value, never the engine's own output.

10.3 The engine contract

The calculation engine is specified as a pure function

$$\texttt{run\_assessment}(\mathit{AssessmentInput},\ \mathit{MethodologyConfig}) \ \longrightarrow\ \mathit{AssessmentResult}$$

with no input or output, no network access, no global mutable state, and no stochastic component. Determinism will therefore be a structural property of the design rather than a behaviour that testing hopes to confirm: there is no source of variation for a test to detect. A determinism test is nonetheless specified, on the principle that a structural guarantee worth relying on is worth verifying.

Each result is to record both the engine version and the methodology version that produced it, and comparisons between results computed under differing methodology versions are to raise a warning rather than being silently rendered side by side.

This contract is a design commitment published in advance of its implementation, which is the point at which it can still be criticised cheaply. A reviewer who believes the contract is too strong, too weak, or unenforceable is invited to say so now.

10.4 Machine-enforced evidence rules

The project's evidence policy is a build gate, not a review convention. Configuration loading fails — and continuous integration fails with it — if any of the following holds.

1. **A typology carries no citation.** The schema requires a non-empty `sources` array on every typology, and each entry must carry both a source key and the finding it supports. An uncited performance value cannot be loaded.
2. **A citation points at an unrecorded source.** Every `sources[].key` appearing anywhere in the configuration tree is collected recursively and checked against the set of keys declared in the bibliography. A citation naming a source the bibliography does not record is a hard error.

This is the rule that makes the traceability claim of Section 2.4 enforceable: a value cannot cite something that does not exist.

3. **The configuration files fall out of version lock-step.** Every file carries a top-level date-stamped `version`, and all must agree. A methodology version moves as a single unit or not at all.
4. **A temperature envelope is inverted or implausible.** The maximum must not fall below the minimum, and no typology may declare a maximum above 3.5 °C. The upper guard encodes a methodology judgment: no retrieved source supports a daytime pedestrian-level air temperature reduction above roughly 3 °C for a single site-level intervention, so a larger value in configuration indicates a calibration error rather than new evidence. Raising the guard requires changing the test, which makes the change visible in review.
5. **The low-confidence declarations are removed.** A test asserts that green façade, rain garden / bioswale, and courtyard greening remain declared as low evidence confidence, guarding against a future edit quietly promoting a weakly evidenced typology.
6. **Default costs appear.** A test asserts that the country defaults ship no cost or emission values pending verification, and that the typology library carries no per-square-metre unit cost. The cost policy of Section 7.10 holds in configuration, not only in prose.

The practical effect is that the integrity claims made in this paper degrade loudly rather than quietly. A contributor who adds a typology without a citation, or who bumps one configuration file without the others, does not produce a subtly wrong tool; they produce a red build.

10.5 Testing and continuous integration

The configuration layer. The test suite asserts that all fourteen typologies load and validate, that every configuration file shares one version stamp, that every typology carries at least one citation with a stated finding, that every source key resolves against the bibliography, that temperature envelopes are ordered and below the 3.5 °C guard, that the three low-confidence typologies remain declared as such, and that no default cost or unverified emission value ships. Negative cases are tested explicitly: an invented citation key, a version mismatch, an uncited typology, and an inverted envelope must each fail the load.

These are the six evidence rules of Section 10.4. Continuous integration runs linting, format checking, strict static type checking, and the full test suite — configuration, scoring, confidence, recommendation, scenarios, determinism — with the 100 % engine coverage gate on every push and pull request, across supported Python versions.

Delivered in Phase 2. Unit tests for every scoring module, with the 100 % engine line-coverage target enforced as a build gate; twenty golden scenarios, each a stored input with a hand-verified expected output and its derivation, serving as regression armour against unintended methodology drift and as the basis of the sensitivity analysis of Section 9.4; and a determinism test asserting byte-identical results across repeated runs and freshly loaded configurations. API contract tests follow with Phase 3.

The ordering is deliberate rather than incidental. The evidence rules were implemented before the scoring code they protect, so that no scoring value can enter the repository uncited at any point in the project's history — not even temporarily, during development.

10.6 Methodology versioning stamped into every result

The methodology version — 2026.08.02 at the time of writing — stamps this document, the Methodology Report in the repository, and every configuration file, and is recorded, together with the engine version, in every assessment result the engine produces. Lock-step across the configuration files is enforced by the build.

This has a consequence that matters for how the tool behaves over time. Existing assessments are *never* to be silently recomputed. A result retains the version that produced it, and the interface indicates when a newer methodology version is available, leaving the decision to re-run with the user. A city whose priority list was produced under version 2026.08.02 can therefore explain, years later, exactly which rules generated it — because those rules are a tagged, public, immutable artefact rather than the current state of a codebase.

10.7 What this arrangement does and does not guarantee

It is worth being precise about the limits of these guarantees, since this section makes the strongest reproducibility claims in the paper.

The arrangement guarantees, today, that every methodology value in the repository carries a citation to a recorded source, that no citation names a source the bibliography does not hold, and that a change to any value is versioned and visible across every file at once. Once the engine exists, it extends to the guarantee that the tool applies exactly the methodology described in this document. These are properties of internal consistency and traceability, not of correctness.

It does *not* guarantee that the methodology is correct, that the adopted envelopes are the right ones, or that the cited sources support the weight placed on them. Those are scientific questions, and no build gate can settle them. What the arrangement does is make them *answerable* by an external reviewer: the complete parameterisation is public, each value names its source, and Appendix E states what was actually verified about each source. The engineering makes the science auditable; it does not make it right.

11 Worked example

This example is illustrative. The site does not exist, and the result is not a finding about any real place.

The calculation was performed **by hand** from the formulas and configuration values specified in this paper, and every number below is also asserted, digit for digit, by the engine's golden-scenario suite — the hand derivation is the expected value, never the engine's own output. Its purpose is to make the specification concrete enough to check: a reader who disagrees with any number below can locate the equation that produced it and recompute it.

One quantity still rests on an assumption: the grid emission factor of Section 7.9, which no shipped configuration supplies. Section 11.7 states how much the conclusion depends on it.

11.1 The illustrative site and intervention

A neighbourhood-scale street corridor in a dense, largely sealed inner-city area in a temperate climate. The proposed intervention is street tree planting (typology 01). Table 12 gives the inputs.

Table 12: Inputs to the illustrative assessment. Fields left unsupplied are shown explicitly, because their absence changes both the result and its confidence.

Group	Field	Value
Project	Assessment scale	Neighbourhood
Site	Site area	6 000 m^2
	Existing tree canopy	8 %
	Existing green cover	12 %
	Impervious surface	85 %
	Soil availability	Limited
	Irrigation availability	Occasional
Climate	Climate zone	Temperate
	Land surface temperature anomaly	4.2 °C
	Solar exposure	High
Vulnerability	Population density	High
	Vulnerable population presence	High
	Access to cooled indoor space	Low
Intervention	Typology	Street tree planting (01)
	New canopy area at maturity	1 200 m^2
Energy	Nearby building cooling demand relevant	Yes
	Annual cooling energy demand	450 000 kW h
	Energy price	*not supplied*
Cost	Capital cost	*not supplied*

11.2 Layer 1 — Baseline assessment

An LST anomaly is available, so the data-rich path of Equation (4) applies. Normalising by Equation (1), $S_{\mathrm{LST}} = \mathrm{clamp}(10 \times 4.2) = 42.0$. The impervious percentage enters directly as $S_{\mathrm{imp}} = 85$; solar exposure *high* normalises to $S_{\mathrm{solar}} = 75$ by Table 2; and the vegetation deficit is $S_{\mathrm{vegdef}} = \mathrm{clamp}(100 - 12) = 88$ by Equation (5). Then

$$\begin{aligned} S_{\mathrm{heat}} &= 0.40(42.0) + 0.25(85) + 0.20(75) + 0.15(88) \\ &= 16.80 + 21.25 + 15.00 + 13.20 \;=\; \mathbf{66.25}. \end{aligned}$$

For vulnerability, population density and vulnerable-group presence are both *high* and normalise to 75. Access to cooled indoor space is *low*, which under the inverted mapping of Section 5.3 yields a cooling access *deficit* of 100. By Equation (6),

$$S_{\text{vuln}} = 0.40(75) + 0.40(75) + 0.20(100) = 30.00 + 30.00 + 20.00 = \mathbf{80.00}.$$

The Heat Priority Index follows from Equation (7):

$$\text{HPI} = 0.60(66.25) + 0.40(80.00) = 39.75 + 32.00 = \mathbf{71.75},$$

which falls in the *High* band of Table 4. Note that vulnerability (80.0) exceeds heat exposure (66.25) here, so the equity-forward weighting of Section 7.12 raises this site's final score relative to a heat-only instrument. That is the disclosed value choice operating as intended, and a reviewer who rejects the choice should read this result as correspondingly inflated.

11.3 Layer 2 — NbS performance

Each of the four adjustment conditions of Table 5 is derived from inputs already supplied.

Canopy. Existing tree canopy is 8 % of 6 000 m^2, or 480 m^2. Adding 1 200 m^2 of new canopy at maturity gives 1 680 m^2, which is 28.0 % of the site — inside the *good* band ($25 \le c < 40$), so $f_{\text{canopy}} = 1.0$. The site falls short of the 40 % inflection identified by Ziter et al. (2019), which is why it does not reach *excellent*.

Soil–water. Soil availability *limited* maps to *moderate*; irrigation availability *occasional* also maps to *moderate*. The more limiting of the two is *moderate*, so $f_{\text{soilwater}} = 0.8$.

Scale. Assessment scale *neighbourhood* maps to *good*, so $f_{\text{scale}} = 1.0$.

Climate. Street tree planting is a *green*-family typology in a *temperate* zone, which Table 6 rates *good*, so $f_{\text{climate}} = 1.0$.

By Equation (8),

$$A = 0.40(1.0) + 0.25(0.8) + 0.20(1.0) + 0.15(1.0) = 0.400 + 0.200 + 0.200 + 0.150 = \mathbf{0.95}.$$

Street tree planting has base cooling score $B = 75$ and envelope $[\tau_{\min}, \tau_{\max}] = [0.5,\ 3.0]$. By Equation (9),

$$S_{\text{cool}} = \text{clamp}(75 \times 0.95) = \mathbf{71.25}.$$

The temperature range follows from Equations (10) and (11), and the clipping rule binds on the lower bound:

$$\Delta T_{\min} = \text{clip}(0.5 \times 0.95,\ 0.5,\ 3.0) = \text{clip}(0.475,\ 0.5,\ 3.0) = \mathbf{0.50}\ ^{\circ}\text{C},$$
$$\Delta T_{\max} = \text{clip}(3.0 \times 0.95,\ 0.5,\ 3.0) = \text{clip}(2.85,\ 0.5,\ 3.0) = \mathbf{2.85}\ ^{\circ}\text{C}.$$

The raw product $0.5 \times 0.95 = 0.475$ falls below the literature floor and is clipped up to it; the upper bound is unaffected because $A < 1$. This is the point marked on Figure 5.

The midpoint is $\overline{\Delta T} = (0.50 + 2.85)/2 = 1.675\,°\mathrm{C}$, which exceeds $1.5\,°\mathrm{C}$ and therefore falls in the *high* heat-index improvement bucket. Shade potential, by Equation (12), is $\mathrm{clamp}(1200/6000 \times 100) = 20.0\,\%$.

11.4 Layer 3 — Impact and feasibility

Energy. All three preconditions of Section 7.8 hold: nearby building cooling demand is confirmed relevant, an annual demand of 450 000 kW h is supplied, and street tree planting is flagged as capable of affecting adjacent building loads. By Equation (17),

$$E_{\min} = 450\,000 \times 0.50 \times 0.02 = 4\,500\,\mathrm{kW\,h/yr},$$
$$E_{\max} = 450\,000 \times 2.85 \times 0.04 = 51\,300\,\mathrm{kW\,h/yr}.$$

The upper bound is roughly eleven times the lower. That width is not a defect of the arithmetic; it is the product of a sixfold temperature envelope and a twofold sensitivity range, and it is the honest consequence of what the evidence supports. A user who needs a tighter figure needs a building energy model.

Greenhouse gases. Version 2026.08.02 ships no country emission factors (Section 7.9). *For this illustration only*, assume the deploying authority has supplied a locally verified grid emission factor of 0.30 kg h/kW. This value is not a Criterra estimate, is not shipped with the tool, and carries no citation; it is a placeholder input chosen to demonstrate Equation (18). On that assumption,

$$G = 1\,350\text{–}15\,390\ \mathrm{kg\ CO_2e\ per\ year, or\ about\ 1.4–15.4\ t\ CO_2e\ per\ year.}$$

Under the shipped configuration with no factor supplied, this output would read *not calculated*.

Costs. No capital cost and no energy price were supplied. Consistent with Section 7.10, capital cost, annual cost savings, simple payback, and cost feasibility are all reported as *not estimated*. Cost feasibility is therefore excluded from the final aggregation and its weight redistributed.

Co-benefits. Street tree planting carries library default levels of *medium* biodiversity, *medium* stormwater, *high* public health, and *high* urban quality. As noted in Section 7.7, the library supplies no default for social inclusion, so it falls to the neutral *unknown* value of 50 and is itemised as an applied assumption. By Equation (14),

$$\begin{aligned} S_{\text{cobenefit}} &= 0.25(50) + 0.25(50) + 0.20(75) + 0.15(50) + 0.15(75) \\ &= 12.50 + 12.50 + 15.00 + 7.50 + 11.25 = \mathbf{58.75}. \end{aligned}$$

NbS Suitability. Applying the rules of Table 7: the site offers sixty times the 100 m^2 minimum ($S_{space} = 100$); soil availability *limited* exactly meets the typology's *limited* requirement ($S_{soil} = 75$); irrigation *occasional* exactly meets the *occasional* requirement ($S_{water} = 75$); maintenance intensity and land use were not supplied, so both take the neutral 50, itemised. Then

$$S_{suit} = 0.30(100) + 0.25(75) + 0.20(75) + 0.15(50) + 0.10(50) = \mathbf{76.25},$$

close to — and slightly below — the 78 assumed in the previous version of this example, because the neutral defaults for maintenance and land use hold the score down where full inputs would not.

11.5 Aggregation

Cost feasibility is unavailable, so Equation (24) applies over the remaining five components, whose nominal weights sum to $\sum_{i \in \mathcal{A}} w_i = 0.90$.

Table 13: Aggregation of the illustrative assessment. Cost feasibility is excluded and its 0.10 weight redistributed proportionally across the remaining five components.

Component	Score	Nominal w_i	Renormalised w_i'	Contribution
Heat Priority Index	71.75	0.25	0.2778	19.93
Cooling Potential	71.25	0.25	0.2778	19.79
NbS Suitability	76.25	0.15	0.1667	12.71
Vulnerability	80.00	0.15	0.1667	13.33
Co-benefits	58.75	0.10	0.1111	6.53
Cost Feasibility	*not estimated — excluded*			—
NbS Cooling Opportunity Score				**72.29**

The NbS Cooling Opportunity Score is **72.3**, which falls in the *Strong opportunity* band (61–80) of Table 4.

11.6 Confidence and reported result

Applying the branched confidence model of Section 8.3 with the field-to-block mapping enumerated at version 2026.08.02: the cooling block has eight of its ten slots supplied (80 % → *high*; street tree planting carries *high* evidence confidence, so no cap applies). The energy block has the relevance confirmation and the demand but no energy price (two of three, 66.7 % → *medium*). The economic block has none of its four fields (*low*). The equity block has three of its six fields (50 % → *medium*). The overall rating is the lower median of {high, medium, low, medium}: *medium*.

[b] On an assumed locally supplied grid emission factor of 0.30 kg h/kW. The shipped configuration would report this as *not calculated*.

Assumptions the engine itemises for this run: maintenance intensity and land use defaulted to the neutral 50 (suitability); the grid emission factor supplied by the user rather than from configuration; energy price absent, so cost savings not computed; the four cited co-benefit library defaults applied;

Table 14: The illustrative assessment as it would be reported, with confidence ratings and applied assumptions. All temperature values are daytime, pedestrian-level air temperature reductions.

Output	Value	Confidence
Heat Priority Index	71.8 — *High*	High
Cooling Potential Score	71.3	High
Temperature reduction	0.50–2.85 °C (daytime, pedestrian-level air)	High
Heat-index improvement	*High* (midpoint 1.675 °C)	High
Shade potential	20.0 % of site	High
Energy savings	4 500–51 300 kW h per year	Medium
GHG avoided	1.4–15.4 t CO_2e per year[b]	Medium
Capital cost	*Not estimated — no cost data supplied*	Low
Simple payback	*Not estimated*	Low
Cost feasibility	*Not estimated — excluded from aggregation*	Low
Co-benefit Score	58.8	Medium
Opportunity Score	**72.3 — *Strong opportunity***	Medium

social inclusion defaulted to the neutral 50 (no library default); public accessibility, safety concern, and community participation defaulted to the neutral 50 (equity); cost feasibility not estimated, excluded from the aggregation with its weight redistributed.

11.7 How much the result depends on the assumed values

At the previous methodology version the NbS Suitability Score was assumed rather than derived, and this subsection quantified the assumption's reach: across the full 0–100 range of the sub-score the Opportunity Score spans 67.9–76.3 and never leaves the *Strong opportunity* band. The sub-score is now derived (Table 7), and its derived value of 76.25 lands within two points of the earlier assumption — the conclusion is unchanged. Category-migration behaviour across the whole scenario set is quantified in Section 9.4.

The one remaining assumed input, the grid emission factor, affects only the greenhouse-gas output, which scales linearly with it and enters no score.

11.8 What this example demonstrates

Four properties of the methodology are visible in the arithmetic above, and each corresponds to a design commitment from Section 2.4.

The clipping rule binds in practice. The raw lower bound $0.5 \times 0.95 = 0.475$ was clipped up to the literature floor. On a better site the rule would bind at the ceiling instead, and the reported range would not exceed 3.0 °C regardless of how favourable the conditions were.

Refusal is a first-class output. Three of the twelve reported quantities are *not estimated*, and a fourth — annual cost savings — is not reported at all for want of an energy price. The tool did not substitute a plausible capital cost, and consequently did not produce a payback figure — the single number most likely to be extracted from a report and quoted without qualification.

Exclusion preserves ordering. Because cost feasibility was excluded rather than set to a neutral

50, this assessment remains comparable to another option for the same site that also lacks cost data, without either being rewarded for the absence.

The ranges are wide, and stay wide. The reported temperature range spans nearly a factor of six and the energy range a factor of eleven. Nothing in the methodology narrows them for presentational comfort.

12 Limitations and responsible use

The limitations below are not a disclaimer appended for legal comfort. Several of them — the daytime restriction, the air-temperature-only scope, the geographic skew of the evidence — materially constrain what the tool's outputs mean, and a user who does not know about them will misread the results. They are stated at length for that reason.

12.1 Screening level

Outputs are indicative estimates produced from simplified typology factors and from user-supplied or default assumptions. They are not microclimate simulation outputs, and they must not be presented as predicted temperatures at a location. The tool has no representation of site geometry, no surface energy balance, no wind field, and no time dimension beyond the daytime/nighttime distinction it declines to model.

The practical consequence is that small differences in an Opportunity Score between two options carry little meaning. The tool distinguishes a strong opportunity from a weak one; it does not rank two strong ones against each other with any reliability. Users comparing closely scored options should treat them as tied and decide on grounds the tool does not represent.

12.2 Daytime only

All cooling values in this methodology are daytime values. The tool produces no nighttime estimate.

This is a substantive restriction, not a scoping convenience, because the evidence indicates that nighttime behaviour differs both in magnitude and in mechanism. Keravec-Balbot et al. (2026) reports substantially smaller nighttime effects — parks at 1.2 °C against 1.3 °C by day, green walls at 1.4 °C against 3.0 °C. Ziter et al. (2019) finds canopy cooling minimal at night while the warming effect of impervious surfaces *persists* overnight, which means that at night the two mechanisms represented in the tool's Heat Exposure Score no longer offset one another.

The importance of this is that nighttime heat is strongly associated with heat-health outcomes: sustained overnight temperatures prevent physiological recovery, and it is that failure to recover which drives excess mortality during heat events. A favourable daytime result from this tool should **not** be read as evidence of overnight heat relief, and an intervention selected on the strength of a daytime score may deliver much less of the health benefit a city is actually seeking.

A user whose primary concern is nighttime heat should treat this tool's output as addressing a related but different question.

12.3 Air temperature only — thermal comfort is not modelled

The tool estimates air temperature reduction. It does not quantify thermal comfort, which is what a person standing in the street actually experiences.

The two diverge most where shade is the dominant mechanism. Shade improves comfort substantially by reducing radiant load, even where the effect on air temperature is small, and comfort indices capture this while air temperature does not. Zölch et al. (2016) reports a 13 % PET reduction from trees at pedestrian level, which illustrates the magnitude of what is omitted; that study also found that placement matters more than quantity, a design insight this methodology cannot express through an air-temperature envelope.

The direction of the resulting error is known and is worth stating plainly: **the tool systematically understates the benefit of shade-dominant interventions.** Street tree planting, shaded pedestrian corridors, and permeable shaded plazas deliver more human benefit than their air-temperature figures convey.

This bias was not corrected, and the decision not to correct it deserves explanation. Applying an uplift to shade-dominant typologies would require a conversion factor between air temperature and comfort that no retrieved source supplies, and inventing one would trade a known, stated, one-directional bias for an unknown one concealed inside a number. A stated bias that users can reason about is preferable to a hidden correction they cannot.

12.4 The evidence base is geographically uneven

The underlying literature is dominated by temperate and subtropical Northern Hemisphere cities. Tropical and arid-climate estimates rest on fewer studies with wider intervals — visible in Figure 1, where the tropical street-tree estimate carries a ±1.6 °C interval and the arid estimate ±1.4 °C on a central value of 1.7 °C, an interval nearly as large as the estimate itself.

The energy sensitivity of Section 7.8 derives largely from North American evidence, and its transferability to other building stocks and climates is untested.

Applications in under-represented contexts should treat outputs as more uncertain than the confidence rating alone suggests. The confidence model of Section 8.3 measures input completeness; it does not measure how well the evidence base covers the user's part of the world. A complete assessment of a site in a tropical city may report high cooling confidence while resting on thinner evidence than the same assessment in a temperate one. This is a known gap in the confidence model and a candidate for a future methodology version.

12.5 Modelling bias is mitigated, not eliminated

Much of the synthesised evidence derives from simulation rather than field measurement, and simulation tends to report larger effects. Section 6.5.1 describes the downward correction applied through conservative bound-setting. That correction is a judgment about direction informed by a single observable ratio; it is not a calibrated adjustment, and it does not eliminate the bias. Adopted values may still be optimistic.

12.6 No default costs

Economic outputs require user-supplied costs (Section 7.10). Absent them, the tool reports no capital cost, no payback, and no cost feasibility, and excludes the latter from aggregation. A user seeking an investment case from the tool alone will not get one. This is deliberate, and the reasoning is given in Section 7.10; but it is a real functional limitation and should be planned for.

12.7 The weights are contestable

The aggregation weights, including the equity-forward emphasis quantified in Section 7.12, are value choices. Users who disagree should adjust them in `config/weights.yaml` and re-run, rather than reinterpreting outputs produced under weights they reject. Reinterpretation is untraceable; reconfiguration is versioned, and the resulting scores carry the version stamp of the configuration that produced them.

12.8 Specification gaps at version 2026.08.02

Three parts of the methodology are declared but not fully specified at this version, and are collected here so that a reviewer has them in one place: the input-to-sub-indicator rules for the NbS Suitability and Equity scores (Section 7.7); the payback bracket boundaries and combination rule for cost feasibility (Equation (21)); and the enumeration of which input fields feed which confidence block (Section 8.3). In addition, the country emission factor table ships empty (Section 7.9), so greenhouse-gas outputs are reported as *not calculated* unless a deployment supplies its own factor. Each of these requires a version bump and a corresponding update to this document when it is resolved.

12.9 Misuse cases

Three specific misuses are anticipated. They are named because each is likely, each is damaging, and each is easy to commit without intending to.

12.9.1 Treating outputs as predicted temperatures

The most likely misuse is quoting a temperature reduction range as a prediction: “this project will reduce temperatures by up to 2.85 °C”. It will not. The range is an envelope of what comparable interventions have delivered in the published literature, narrowed by a coarse site adjustment. It carries no location, no time of day beyond “daytime”, no meteorological condition, and no confidence interval in the statistical sense.

The correct formulation names the metric, the range, and the basis: “a screening assessment indicates an indicative daytime pedestrian-level air temperature reduction in the range 0.5–2.85 °C, based on published evidence for this intervention type adjusted for site conditions”. This is longer, and it is what the tool actually supports.

The failure mode is particularly acute where a range is collapsed to its upper bound for a headline. The upper bound of an envelope describes a well-executed instance under favourable conditions; it is not the expected outcome.

12.9.2 Quoting scores without their confidence

An Opportunity Score of 72.6 at medium confidence and an Opportunity Score of 72.6 at low confidence are different results, and the difference is not cosmetic — the second may rest largely on the neutral defaults of Section 5.4 rather than on anything the user knew about the site.

Scores extracted into a slide, a table, or a funding proposal without their confidence rating and their methodology version have been stripped of the information needed to judge them. The tool attaches confidence wherever a score appears, and reports circulated onward should preserve that pairing. Where a score is quoted, the methodology version should be quoted with it, because the same site may score differently under a later version.

12.9.3 Using the tool to justify a decision already taken

The most consequential misuse is the least visible one. Because the tool is transparent, configurable, and produces a defensible-looking number, it can be run after a decision has been made in order to furnish that decision with evidence — by selecting the option already chosen, by adjusting weights until the preferred site ranks first, or by presenting a single assessment without the alternatives that were considered.

Nothing in the software can prevent this, and no methodology can. What the design does is leave traces: the methodology version is stamped on every result, the weights used are recorded in configuration, applied defaults are itemised, and comparison of options for the same site is a first-class feature rather than an afterthought. An assessment presented without its alternatives, or produced under non-default weights that are not disclosed, should be questioned on those grounds.

The tool is built to compare options and set priorities. Used to ratify a conclusion reached elsewhere, it lends the appearance of rigour to a process that had none, which is worse than having no tool at all.

12.10 Responsible use in summary

- Use the tool to compare options and set priorities, not to justify a decision already taken.
- Combine results with local knowledge and stakeholder input; the tool represents neither.
- Validate assumptions through site visits before committing resources.
- Re-run assessments as better data becomes available, and expect the result to change.
- Report the confidence rating and the methodology version alongside any score that is quoted.
- Name the metric whenever a temperature is reported: daytime, pedestrian-level, air temperature.

13 Governance and methodology evolution

13.1 Versioning

The methodology version — 2026.08.02 — stamps this document, the Methodology Report in the repository, and every configuration file, and is to be recorded in every assessment result the engine produces. The version is a date stamp rather than a semantic version, because a methodology does not have a meaningful notion of a backward-compatible change: any alteration to a value may alter a ranking.

A change to any methodology value requires a version bump and a corresponding update to this document in the same change set. Continuous integration enforces the mechanical half of this — version lock-step across configuration files, and citations for every performance value (Section 10.4) — while the substantive half, that the documentation actually describes the new value, is enforced in review.

Existing assessments are never to be silently recomputed: a result retains the version that produced it, the interface indicates when a newer methodology version is available, leaving the decision to re-run with the user, and comparisons between results produced under differing versions raise a warning. None of this is yet exercised, since no assessment result exists at version 2026.08.02 (Section 10.2); it is the behaviour the engine is specified to have.

13.2 The public change process

Methodology changes are proposed as public pull requests, accompanied by the evidence supporting them. A proposal that changes a quantitative value is expected to state which source supports the new value, what was verified about that source and how, and why the existing value should be displaced. The verification standard is the one described in Appendix E: bibliographic details checked against the publisher record, an institutional repository, or an indexing service, with the verification level recorded honestly rather than overstated.

Because the methodology lives in configuration rather than in code, the diff of such a proposal is legible to a domain expert who does not read the implementation language. This is a deliberate property. A reviewer evaluating a change to the green façade envelope should be able to see a temperature range, a confidence rating, and a citation change, and nothing else.

13.3 How to challenge this methodology

Methodology critique is a first-class contribution to this project, not an inconvenience to be managed. The route is to open an issue in the repository at `https://github.com/Dimitrios-Kafetzis/CriterraNatureCoolingTool`, referencing the section number of this document and, where possible, the literature supporting the alternative position.

Four areas are in particular need of external scrutiny, and are listed in descending order of how much a change would affect the tool's outputs.

1. **The aggregation weights** (Section 7.11), and specifically the equity-forward double contribution of vulnerability (Section 7.12). This is the tool's most consequential unsourced choice and a value position rather than a finding. The sensitivity analysis of Section 9.4 quantifies its influence: rank orderings prove substantially stable under single-weight perturbation, which bounds — but does not settle — the dispute.
2. **The green façade and rain garden / bioswale envelopes** (Section 6.5.3). For green façades the sources conflict outright and the tool adopts a conservative resolution based on a hypothesis about measurement position that neither source states. For rain gardens and bioswales no

source quantifies air-temperature cooling at all, and the adopted range is reasoned from adjacent evidence. Both are candidates for replacement by better evidence, and both would benefit from a reviewer who knows this literature better than the derivation in Appendix A demonstrates.

3. **The transferability of the energy sensitivity** (Section 7.8). The 2–4 % per °C figure is applied globally from predominantly North American evidence. Reviewers with access to equivalent evidence for other building stocks and climates would materially improve this part of the methodology, and evidence that the figure does *not* transfer would be equally valuable.
4. **The downward correction for modelling bias** (Section 6.5.1). The correction rests on a single observable ratio for a single measure, used to justify a direction rather than a magnitude. A reviewer who considers the resulting envelopes too conservative — or not conservative enough — is disputing exactly the right thing.

13.4 What a good challenge looks like

The most useful critiques share a shape. They identify a specific value or rule, by section number; they state what is wrong with it, distinguishing between an arithmetic error, a misread source, a source that does not support the weight placed on it, and a value judgment the critic rejects; and where the critique is evidential rather than a difference of values, they name a source and what it actually reports, including which metric it reports it in.

The last point is the one most often missed. A critique of a cooling envelope that cites a figure without stating whether it is air temperature, land surface temperature, or a comfort index cannot be acted on, because the methodology cannot adopt a value whose metric is unknown (Section 6.2). This is the same standard the methodology holds itself to.

Critiques that reject a value judgment — the equity weighting being the obvious case — are welcome and will be recorded, but they will generally be resolved by making the configuration easier to rebalance and the consequences easier to see, rather than by replacing one contested judgment with another.

14 Conclusion

This paper has specified, in full, the methodology of the Nature for Cooling Rapid Assessment Tool at methodology version 2026.08.02: every input and its normalisation, the fourteen-typology evidence library and the derivations behind it, every scoring formula and every weight, the treatment of uncertainty and missing data, the design of the sensitivity analysis, and the engineering arrangement that keeps the software in correspondence with the specification.

The instrument occupies a deliberately narrow position. It is faster and more comparable than unstructured judgment, and far less capable than microclimate simulation. It exists for the triage step: narrowing a long list of candidate sites and interventions to a short one, on a transparent and evidence-referenced basis, and recording why. Judged as a simulation it fails; judged as a screening instrument, the questions that matter are whether its values are traceable, whether its uncertainties are honestly represented, whether its rankings are stable, whether its value judgments are disclosed, and whether it declines to answer when its inputs do not support an answer (Section 3.5).

Several of the methodology's choices lower its apparent authority, and they were the substance of the work rather than concessions to it. Adopted cooling envelopes sit at or below synthesis central estimates because much of the underlying evidence is modelling-derived and modelling reports larger effects than measurement. The mixed intervention package is capped at the best-evidenced single-measure ceiling because no retrieved source quantifies super-additive cooling. Reported temperature ranges are clipped to the literature envelope, so that a favourable site raises a relative score but never a physical claim. Per-typology energy factors that no source supported were deleted and replaced by a derivation from a published sensitivity. No default unit costs ship, so the tool declines to produce the payback figure that would otherwise be its most quotable output. Green façades, rain gardens, and courtyard greening are labelled low confidence and their confidence caps propagate into results.

The limitations are correspondingly explicit. All values are daytime, and nighttime heat — the more consequential quantity for health outcomes — is not estimated. The quantity modelled is air temperature, not thermal comfort, which means the tool systematically understates shade-dominant interventions; the bias is stated rather than corrected by an unsourced factor. The evidence base is skewed toward temperate and subtropical Northern Hemisphere cities, and the confidence model does not yet measure that skew. The aggregation weights are expert judgment, including a disclosed equity-forward stance that gives vulnerability an effective weight of 0.25 against heat exposure at 0.15. The sensitivity analysis of Section 9.4 finds the ranking substantially stable under ±25 % single-weight perturbation — worst-case pairwise stability 0.973 7, with three near-boundary category changes in 240 scenario-perturbation combinations — which bounds how much the contestable weights actually move decisions.

The methodology was first offered for review with its gaps disclosed rather than closed, because the values most in need of scrutiny are settled most cheaply before an engine, a user base, and a set of published assessments have accumulated around them. Version 2026.08.02 closes the disclosed specification gaps — the sub-indicator rules, the cost-feasibility derivation, and the confidence field mapping — without altering any evidence-derived value. The tool, its complete configuration, and this document are public under the Apache License 2.0. Every value it applies names its source; Appendix E states what was actually verified about each of those sources, including where publisher restrictions meant a finding was confirmed from an abstract or an indexing record rather than a full text. That disclosure is not a weakness in the evidence base. It is the property that makes the evidence base checkable, and it is what the authors would want to know before trusting an instrument of this kind.

References

Akbari, H., M. Pomerantz, and H. Taha (2001). "Cool surfaces and shade trees to reduce energy use and improve air quality in urban areas". In: *Solar Energy* 70.3, pp. 295–310. DOI: 10.1016/S0038-092X(00)00089-X.

Bowler, D. E., L. Buyung-Ali, T. M. Knight, and A. S. Pullin (2010). "Urban greening to cool towns and cities: A systematic review of the empirical evidence". In: *Landscape and Urban Planning* 97.3, pp. 147–155. DOI: 10.1016/j.landurbplan.2010.05.006.

Ember (2026). *Yearly Electricity Data / Electricity Data Explorer*. Electricity carbon intensity for 215 countries; released under CC-BY-4.0. Ember Energy. URL: https://ember-energy.org/data/yearly-electricity-data/ (visited on 07/30/2026).

Gunawardena, K. R., M. J. Wells, and T. Kershaw (2017). "Utilising green and bluespace to mitigate urban heat island intensity". In: *Science of the Total Environment* 584–585, pp. 1040–1055. DOI: 10.1016/j.scitotenv.2017.01.158.

Keravec-Balbot, T., R. Schoetter, and V. Masson (2026). "Adapting cities to heat: an umbrella review of the effectiveness of urban cooling measures across climates". In: *Environmental Research Letters* 21.14, p. 143003. DOI: 10.1088/1748-9326/ae870e.

Nardo, M., M. Saisana, A. Saltelli, S. Tarantola, A. Hoffmann, and E. Giovannini (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing / European Commission Joint Research Centre. ISBN: 978-92-64-04345-9. URL: https://publications.jrc.ec.europa.eu/repository/handle/JRC47008.

Norton, B. A., A. M. Coutts, S. J. Livesley, R. J. Harris, A. M. Hunter, and N. S. G. Williams (2015). "Planning for cooler cities: A framework to prioritise green infrastructure to mitigate high temperatures in urban landscapes". In: *Landscape and Urban Planning* 134, pp. 127–138. DOI: 10.1016/j.landurbplan.2014.10.018.

Oke, T. R. (1976). "The distinction between canopy and boundary-layer urban heat islands". In: *Atmosphere* 14.4, pp. 268–277. DOI: 10.1080/00046973.1976.9648422.

Rahman, M. A. et al. (2020). "Tree cooling effects and human thermal comfort under contrasting species and sites". In: *Agricultural and Forest Meteorology* 287, p. 107947. DOI: 10.1016/j.agrformet.2020.107947.

Reid, C. E., M. S. O'Neill, C. J. Gronlund, S. J. Brines, D. G. Brown, A. V. Diez-Roux, and J. Schwartz (2009). "Mapping community determinants of heat vulnerability". In: *Environmental Health Perspectives* 117.11, pp. 1730–1736. DOI: 10.1289/ehp.0900683.

Santamouris, M. (2014). "Cooling the cities – A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments". In: *Solar Energy* 103, pp. 682–703. DOI: 10.1016/j.solener.2012.07.003.

Venter, Z. S., T. Chakraborty, and X. Lee (2021). "Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms". In: *Science Advances* 7.22, eabb9569. DOI: 10.1126/sciadv.abb9569.

World Bank (2021). *A Catalogue of Nature-based Solutions for Urban Resilience*. Report. Global Program on Nature-based Solutions for Climate Resilience and City Resilience Program. Washington, D.C.: World Bank Group. URL: https://documents.worldbank.org/en/publication/documents-reports/documentdetail/502101636360985715/.

Ziter, C. D., E. J. Pedersen, C. J. Kucharik, and M. G. Turner (2019). "Scale-dependent interactions between tree canopy cover and impervious surfaces reduce daytime urban heat during summer". In: *Proceedings of the National Academy of Sciences* 116.15, pp. 7575–7580. DOI: 10.1073/pnas.1817561116.

Zölch, T., J. Maderspacher, C. Wamsler, and S. Pauleit (2016). "Using green infrastructure for urban climate-proofing: An evaluation of heat mitigation measures at the micro-scale". In: *Urban Forestry & Urban Greening* 20, pp. 305–316. DOI: 10.1016/j.ufug.2016.09.011.

A The complete NbS typology library

This appendix reproduces the full typology library at methodology version 2026.08.02, as it appears in `config/nbs_typologies.yaml`. All temperature values are daytime, pedestrian-level *air* temperature reductions in degrees Celsius, relative to an equivalent unimproved site.

A.1 Adopted values and suitability conditions

Table 15: Adopted values and suitability conditions for all fourteen typologies. *Bldg. energy* indicates whether the typology is flagged as capable of affecting adjacent building cooling loads, a precondition for the energy derivation of Section 7.8.

ID	Typology	Base score	Envelope (°C)	Conf.	Bldg. energy	Min. area (m^2)	Soil / water required
Green							
01	Street tree planting	75	0.5–3.0	High	Yes	100	Limited / occas.
02	Shaded pedestrian corridor	80	0.5–3.0	Medium	No	200	Limited / occas.
03	Pocket park	65	0.3–1.5	High	No	200	Limited / occas.
04	Urban forest / dense canopy	90	1.0–3.0	High	Yes	2000	Moderate / occas.
05	Park upgrade	70	0.5–2.0	High	No	500	Limited / occas.
06	Schoolyard greening	75	0.5–2.0	Medium	Yes	300	Limited / occas.
Building							
07	Green roof	45	0.1–1.0	Medium	Yes	50	None / occas.
08	Green façade	50	0.3–2.0	Low	Yes	20	None / reliable
Blue-green							
09	Rain garden / bioswale	55	0.1–0.8	Low	No	50	Moderate / none
10	Blue-green corridor	85	1.0–3.0	Medium	No	1000	Moderate / none
11	Riparian restoration	85	1.0–3.0	Medium	No	1000	Moderate / none
Hybrid							
12	Permeable shaded plaza	70	0.5–2.5	Medium	No	200	Limited / occas.
13	Courtyard greening	60	0.3–1.5	Low	Yes	50	Limited / occas.
14	Mixed NbS package	80	1.0–3.0	Medium	Yes	500	Limited / occas.

No typology declares an unsuitable climate zone at version 2026.08.02. Climate influences performance through the adjustment matrix of Table 6 rather than through outright disqualification.

A.2 Primary cooling mechanisms and co-benefit defaults

Table 16: Primary cooling mechanism and default co-benefit levels per typology. The library supplies no default for the *social inclusion* sub-indicator of Equation (14); absent a user override it falls to the neutral *unknown* value and is itemised as an applied assumption.

ID	Primary cooling mechanism	Biodiv.	Storm.	Health	Urban q.
01	Shade and evapotranspiration	Med	Med	High	High
02	Continuous shade and evapotranspiration	Med	Low	High	High
03	Evapotranspiration and shade	Med	Med	High	High
04	Dense shade and evapotranspiration	High	High	High	High
05	Increased canopy and evapotranspiration	High	Med	High	High
06	Shade and evapotranspiration	Med	Med	High	High
07	Roof surface cooling and evapotranspiration	Med	High	Low	Med
08	Wall shading and evapotranspiration	Low	Low	Low	Med
09	Evapotranspiration and infiltration	Med	High	Low	Med
10	Evaporation from water surfaces and canopy shade	High	High	High	High
11	Evaporation from water surfaces and riparian canopy	High	High	Med	High
12	Shade, surface permeability and evaporation	Low	High	Med	High
13	Shade and evapotranspiration in enclosed space	Med	Med	Med	High
14	Combined shade, evapotranspiration and evaporation	High	High	High	High

A.3 Citations and the findings they support

Every performance value in the library carries at least one citation, and each citation records the specific finding it supports. Configuration loading fails if a typology is uncited or if a citation names a source absent from the bibliography (Section 10.4). The findings below are reproduced from the configuration.

Table 17: Citations supporting each typology's adopted values, with the finding each citation supports. Metrics are stated where a source reports something other than air temperature.

ID	Source	Finding relied upon
01	Keravec-Balbot et al. 2026	Street trees 1.5 °C central estimate for pedestrian-level daytime air temperature; (4.0 ± 1.6) °C in tropical (A) and (1.7 ± 1.4) °C in arid (B) climates.
	Ziter et al. 2019	Air temperature decreases nonlinearly with canopy cover, greatest cooling above about 40 % cover; peak effect at 60–90 m scale.
	Bowler et al. 2010	Empirical studies broadly support cooler air beneath tree canopy.
	Rahman et al. 2020	Species and leaf area index materially change delivered cooling.
02	Keravec-Balbot et al. 2026	Street trees 1.5 °C and shading devices 2.0 °C pedestrian-level daytime air temperature reduction.
	Zölch et al. 2016	Shade placement in heat-exposed locations outperforms maximising total green cover; trees −13 % PET *(comfort index)*.
03	Bowler et al. 2010	Meta-analysis: park on average 0.94 °C cooler by day; larger parks and those with trees cooler.
	Keravec-Balbot et al. 2026	Parks 1.3 °C pedestrian-level daytime air temperature reduction.
04	Keravec-Balbot et al. 2026	Urban forests 2.1 °C, among the highest-performing green measures.
	Ziter et al. 2019	Greatest cooling where canopy cover exceeds 40 %; effect grows with patch scale to 60–90 m.
	Bowler et al. 2010	Larger, tree-rich green sites cooler during the day.
05	Bowler et al. 2010	Park mean 0.94 °C daytime cooling; parks with trees cooler than open turf.
	Keravec-Balbot et al. 2026	Parks 1.3 °C, green areas 2.0 °C.
06	Keravec-Balbot et al. 2026	Green areas 2.0 °C and street trees 1.5 °C.
	Ziter et al. 2019	Cooling maximised at city-block scale (60–90 m), comparable to a schoolyard footprint.
	Bowler et al. 2010	Tree-containing green sites outperform open green space.

continued on next page

Table 17 continued

ID	Source	Finding relied upon
07	Keravec-Balbot et al. 2026	Green roofs 1.1 °C, the lowest-performing green measure in the synthesis.
	Zölch et al. 2016	Green roof effects at pedestrian level negligible *(PET)*.
	Santamouris 2014	0.3–3 K average urban ambient reduction, but *only for city-scale deployment in simulation studies*, not single-building installation.
08	Keravec-Balbot et al. 2026	Green walls 3.0 °C central, with fivefold spread across climate subzones (Cwa 5.7, BWh 5.5, Csa 2.9, Cfa/Cfb 1.3, Af 1.0 °C).
	Zölch et al. 2016	Green façades −5–−10 % PET, below trees at −13 % *(comfort index)*.
09	Keravec-Balbot et al. 2026	No bioswale-specific estimate available; neighbouring categories (green areas 2.0 °C, water bodies 2.1 °C) are not transferable to small drainage features.
	Gunawardena et al. 2017	Tree-dominated greenspace delivers heat-stress relief; small blue features are not identified as significant coolers.
10	Keravec-Balbot et al. 2026	Water bodies 2.1 °C and urban forests 2.1 °C; water bodies and parks reported as largely climate-independent.
	Gunawardena et al. 2017	Green and blue features together offer synergistic benefits; poorly designed bluespace may exacerbate heat stress in humid conditions.
11	Keravec-Balbot et al. 2026	Water bodies 2.1 °C, climate-robust; urban forests 2.1 °C.
	Gunawardena et al. 2017	Bluespace cooling mechanisms; humid-condition heat-stress caveat.
12	Keravec-Balbot et al. 2026	Permeable pavements 2.9 °C, cool pavements 2.0 °C, shading devices 2.0 °C.
	Zölch et al. 2016	Shade positioned in heat-exposed locations is the decisive design factor.
13	Keravec-Balbot et al. 2026	Green areas 2.0 °C and green walls 3.0 °C; enclosed-courtyard geometry not separately resolved.
	Ziter et al. 2019	Cooling scales with patch size; courtyards fall below the 60–90 m optimum.

continued on next page

Table 17 continued

ID	Source	Finding relied upon
	Zölch et al. 2016	Micro-scale greening in dense residential fabric reduces PET; placement dominates quantity *(comfort index)*.
14	Keravec-Balbot et al. 2026	Best-performing individual NbS measures cluster at 2.0–3.0 °C; no synthesis estimate exists for combined packages.
	Gunawardena et al. 2017	Combined green and blue features offer synergistic ecosystem benefits, stated qualitatively and not quantified.

A.4 Calibration notes recorded in configuration

Table 18: Calibration notes carried in the typology library, explaining why an adopted value departs from a naive reading of its sources. Typologies 04, 10, and 11 carry no note; their values follow directly from converging sources.

ID	Note
01	Upper bound capped at 3.0 °C rather than the tropical central estimate of 4.0 °C, which carries a ± 1.6 °C interval and reflects favourable conditions.
02	Scored above street trees because shade is placed where exposure occurs. The narrow-street effect reported in the synthesis is an urban-form effect, not an NbS intervention, and is deliberately not inherited.
03	Explicitly the small end of the park distribution, bounded below the general park estimate by the size relationship reported in Bowler et al. (2010).
05	Represents added cooling over an already partly green baseline, so the achievable increment is bounded below new-forest values.
06	No schoolyard-specific synthesis found; values inherited from small green areas with trees at matching spatial scale. The typology's distinct value is exposure of a vulnerable group, expressed through vulnerability and equity scoring rather than through inflated cooling.
07	Benefits are concentrated at roof level and inside the building beneath. The tool states in its output that green-roof benefits are principally building-level rather than street-level.
08	Sources conflict outright: the synthesis ranks green walls near the top on air temperature while the micro-scale comparison ranks them below trees on comfort, plausibly explained by near-wall measurement position. Low confidence, conservative envelope, explicit caveat in tool output.

continued on next page

ID	Note
09	No source quantifies air-temperature cooling for rain gardens or bioswales specifically. Value is a reasoned lower bound from adjacent evidence. Primary benefits are stormwater and biodiversity, not cooling.
12	The permeable-pavement estimate rests on a narrower evidence base than the vegetation figures, so the adopted upper bound is set below it.
13	Small, enclosed, often sheltered from ventilation. Bounded by scale limitation; no courtyard-specific synthesis exists.
14	The draft methodology's 3.5 °C upper bound was rejected: no retrieved source quantifies super-additive cooling from combining measures. Capped at the best-evidenced single-measure ceiling. The package's advantage is expressed as co-benefit breadth and robustness, not additional degrees.

B Complete weight tables

All weights are from `config/weights.yaml` at version 2026.08.02. Within each block the weights sum to unity. All are expert judgment in the sense of Nardo et al. (2008); where a source informs the *selection* of indicators or the *relative ordering* of weights, it is named.

B.1 Heat Exposure Score (data-rich path only)

Component	Weight
Land surface temperature anomaly	0.40
Imperviousness	0.25
Solar exposure	0.20
Vegetation deficit	0.15
Total	**1.00**

Basis. Component selection reflects established surface-energy-balance drivers of urban heat. Ziter et al. (2019) gives direct empirical support for two components: impervious cover raises air temperature approximately linearly and canopy cover reduces it nonlinearly. The relative weights are expert judgment.

B.2 Vulnerability Score

Component	Weight
Population density	0.40
Vulnerable groups	0.40
Cooling access deficit	0.20
Total	**1.00**

Basis. Indicator selection follows Reid et al. (2009), whose factor analysis of ten heat-vulnerability variables found four factors explaining over 75 % of variance, with air-conditioning prevalence emerging as an independent dimension. Density and vulnerable-group presence are weighted equally because that analysis gives no basis for ranking one above the other; cooling access carries less weight as a single, frequently unknown indicator.

B.3 Heat Priority Index and site adjustment

Heat Priority Index		Site adjustment	
Heat exposure	0.60	Canopy	0.40
Vulnerability	0.40	Soil–water	0.25
		Scale	0.20
		Climate	0.15
Total	**1.00**	**Total**	**1.00**

Basis for site adjustment. Canopy carries the largest weight as the condition with the strongest direct empirical support (Ziter et al. 2019). The climate component is justified by Keravec-Balbot et al. (2026): adding Köppen–Geiger subzone raises variance explained in cooling effectiveness from 12 % to 78 %.

B.4 Composite sub-scores

NbS Suitability		**Co-benefits**		**Equity**	
Space	0.30	Biodiversity	0.25	Vulnerable user benefit	0.35
Soil	0.25	Stormwater	0.25	Public accessibility	0.25
Water	0.20	Public health	0.20	Safety and comfort	0.20
Maintenance	0.15	Social inclusion	0.15	Participation relevance	0.20
Urban context	0.10	Urban quality	0.15		
Total	**1.00**	**Total**	**1.00**	**Total**	**1.00**

The rules mapping user inputs to these individual sub-indicator scores are fixed at version 2026.08.02 in `config/derived_scores.yaml` and reproduced in Table 7. The Equity Score does not enter the final aggregation (Section 7.7). The derived cost feasibility score combines the payback bracket with two delivery-risk modifiers (Equation (22)):

Cost feasibility component	**Weight**
Payback bracket score	0.50
Implementation complexity (inverted)	0.25
Maintenance intensity (inverted)	0.25
Total	**1.00**

B.5 Final aggregation and effective weights

Component	**Nominal**	**Effective**
Heat Priority Index	0.25	—
heat exposure		0.15
vulnerability		0.10
Cooling Potential	0.25	0.25
NbS Suitability	0.15	0.15
Vulnerability (direct)	0.15	0.15
Co-benefits	0.10	0.10
Cost Feasibility	0.10	0.10
Total	**1.00**	**1.00**
Restated, not added:		
Vulnerability, total		**0.25**
Heat exposure, total		**0.15**

The equity-forward double contribution of vulnerability is deliberate and is argued in Section 7.12. The sensitivity-analysis perturbation parameter is ± 25 % (Section 9).

C Input field reference

The definitive enumeration is fixed by the implemented input schema (`nature_cooling.engine.models.Assessme` Fields marked *Required* are exactly those without which a documented formula cannot be evaluated at all; every other field is *Optional*, and its absence is handled through a disclosed default or an explicit *not calculated* status, itemised in the result.

Table 19: Input fields, their status, and the normalisation applied to each.

Field	Status	Normalisation / use
Site characteristics		
`site_area_m2`	Required	Denominator of the canopy condition and of shade potential (Equation (12)).
`existing_tree_canopy_percent`	Optional	Canopy condition derivation (Table 5).
`existing_green_cover_percent`	Optional	Vegetation deficit, clamp$(100 - g)$ (Equation (5)).
`impervious_surface_percent`	Optional	Percentage used directly as a 0–100 score in Equation (4); declared in configuration at version 2026.08.02.
`soil_availability`	Optional	none 25 / limited 50 / moderate 75 / high 100; unknown 50. Also feeds the soil–water condition.
`irrigation_availability`	Optional	none 25 / occasional 50 / reliable 100; unknown 50. Also feeds the soil–water condition.
`current_shade_level`	Optional	very low 25 / low 50 / medium 75 / high 100; unknown 50. Counts toward cooling-block completeness; enters no formula at version 2026.08.02.
`land_use`	Optional	Urban-context suitability sub-indicator: in the typology's typical-use contexts 100, outside 25, unknown 50 (Table 7).
Project information		
`assessment_scale`	Required	Scale condition: city, district → excellent; neighbourhood → good; site, building → moderate.

continued on next page

Table 19 continued

Field	Status	Normalisation / use
`country`	Optional	Grid emission factor lookup in `country_defaults.yaml` (Section 7.9).
Climate and heat exposure		
`climate_zone`	Required	Climate condition lookup (Table 6).
`lst_anomaly_c`	Optional	Linear, 0 °C→0, 10 °C→100, clamped (Equation (1)). Selects the data-rich heat exposure path.
`heat_exposure_level`	Optional	Standard levels. Used directly as the Heat Exposure Score on the data-poor path.
`solar_exposure`	Optional	Standard levels.
`heat_index_concern`	Optional	Standard levels.
Vulnerability and equity		
`population_density`	Optional	Standard levels.
`vulnerable_population_presence`	Optional	Standard levels.
`access_to_cooled_indoor_space`	Optional	**Inverted** levels: high 25 / medium 50 / low 100; unknown 50 (Section 5.3).
`safety_concern`	Optional	Standard levels; equity safety-and-comfort sub-indicator (deficit reading).
`public_accessibility`	Optional	Standard levels; equity sub-indicator. New at version 2026.08.02.
`community_participation`	Optional	Standard levels; equity participation-relevance sub-indicator. New at version 2026.08.02.
NbS intervention		
`nbs_type`	Required	Selects the typology and its climate-family row.
`new_canopy_area_at_maturity_m2`	Optional	Canopy condition and shade potential.

continued on next page

Table 19 continued

Field	Status	Normalisation / use
`expected_maturity_period_years`	Optional	Time-to-benefit class (immediate / short / medium / long term). No maturity discount is applied to cooling values.
`intervention_area_m2`	Optional	Validation warning when exceeding the site area (OQ-04); enters no score.
`implementation_complexity`	Optional	Inverted levels; cost feasibility (Equation (22)) and investment readiness.
`maintenance_intensity`	Optional	Inverted levels; cost feasibility and the suitability maintenance sub-indicator.
Cost and energy (all optional)		
`nearby_building_cooling_demand_relevant`	Optional	Precondition for the energy derivation (Section 7.8).
`annual_cooling_energy_demand_kwh`	Optional	D in Equation (16). Absent: `missing_energy_demand`.
`energy_price_per_kwh`	Optional	p in Equation (19). Absent: cost savings not calculated.
`capital_cost`	Optional	C_{capital} in Equation (20). Absent: capital cost, payback, cost feasibility, and investment readiness all *not estimated*.
`currency`	Optional	ISO 4217 code, echoed with cost outputs; never used in scoring.
`grid_emission_factor_kgco2e_per_kwh`	Optional	ϕ in Equation (18); a user-supplied factor takes precedence over the country table and is itemised as an applied assumption.

D Glossary

Air temperature
The temperature of the air, here measured or estimated at pedestrian level. The quantity this methodology reports. Distinct from land surface temperature and from thermal comfort indices.

Base cooling score
A 0–100 index expressing a typology's relative expected cooling performance before site adjustment. Not a temperature.

Block confidence
A property of an *assessment*: how completely the user supplied the inputs a given output block depends on (Table 9). Distinct from evidence confidence.

Boundary-layer urban heat island
The heat island of the air mass above roof level, distinguished from the canopy-layer and surface heat islands by Oke (1976).

Canopy-layer urban heat island (CUHI)
The heat island of the air between the ground and roof level — the layer people occupy. The quantity relevant to this methodology.

Clamp
$\mathrm{clamp}(x)$ bounds a value to the 0–100 reporting scale.

Clip
$\mathrm{clip}(x,\ l,\ u)$ bounds a value to a stated interval. Used to confine reported temperature estimates to the literature envelope (Section 7.6).

Composite indicator
An index formed by normalising and aggregating several sub-indicators. Construction here follows Nardo et al. (2008).

Cooling Potential Score
The Layer 2 headline score: the typology's base cooling score scaled by the site adjustment factor (Equation (9)).

Envelope
The literature-supported interval of temperature reduction for a typology. Its lower bound represents a poorly performing instance, its upper bound a well-executed instance under favourable conditions. Not a prediction and not a confidence interval.

Evidence confidence
A property of a *typology*: how well grounded its adopted values are (high, medium, low). Propagates into an assessment by capping cooling-block confidence (Section 8.3.1).

Golden scenario
A stored input case with a hand-verified expected output, used as regression armour against unintended methodology drift and as the basis of the sensitivity analysis.

Heat Priority Index (HPI)
The Layer 1 headline score, combining heat exposure and vulnerability (Equation (7)). Independent of the proposed intervention.

Köppen–Geiger classification
A climate classification system whose subzones (Af, BWh, Csa, Cfa, Cfb, Cwa, . . .) raise the variance explained in cooling effectiveness from 12 % to 78 % (Keravec-Balbot et al. 2026).

Land surface temperature (LST)
The radiometric temperature of the ground and building surfaces, typically satellite-derived. Substantially exceeds air temperature (Venter et al. 2021); used here only as a relative screening proxy (Section 5.5).

NbS Cooling Opportunity Score
The final aggregated score (Equation (23)), on a 0–100 scale, for comparing options and sites.

Nature-based solutions (NbS)
Interventions using vegetation, water, and natural processes to deliver benefits — here, urban cooling.

PET (physiologically equivalent temperature)
A thermal comfort index representing perceived thermal conditions. **Not** air temperature; used in this methodology only for relative ranking and caveats.

Screening level
An assessment tier producing indicative, comparable estimates for triage, positioned between unstructured judgment and detailed simulation.

Site adjustment factor
The composite multiplier $A \in [0.5, 1.2]$ derived from canopy, soil–water, scale, and climate conditions (Equation (8)).

Super-additivity
The hypothesis that combined interventions cool more than their strongest component. Not adopted, because no retrieved source quantifies it (Section 6.5.2).

Surface urban heat island (SUHI)
The heat island measured in surface temperatures. Reported at roughly six times the canopy-layer magnitude by Venter et al. (2021).

UTCI (universal thermal climate index)
A thermal comfort index. As with PET, not air temperature, and not reported by this tool.

E Source verification status

Every source enters the methodology's bibliography only after its bibliographic details have been checked against the publisher record, an institutional repository, or an indexing service. Table 20 reproduces the verification register from `docs/methodology/BIBLIOGRAPHY.md` at version 2026.08.02.

Three verification levels are used.

Full Bibliographic details *and* the specific quantitative finding used by the methodology were both confirmed from a retrieved source.

Metadata + finding (secondary) Bibliographic details confirmed from the publisher or repository; the quantitative finding confirmed from an authoritative secondary description — a publisher abstract page or an indexing service — rather than from the full text.

Metadata only Bibliographic details confirmed; the source is referenced qualitatively only, and no numeric value in the tool depends on it.

This register is reproduced rather than summarised because the distinction it records is material. Several publisher sites — Elsevier, PNAS, Science — block automated retrieval, so a number of findings were confirmed from open repositories, PubMed, or publisher abstract pages. Where a value in the tool depends on a finding that is not *Full*, the configuration marks its confidence accordingly.

A reader should not read *metadata + finding (secondary)* as an implied full-text reading. It means what it says: the number was confirmed from an authoritative description of the paper, not from the paper.

Table 20: Verification status of every source cited by the methodology.

Key	Verified	Basis and notes
`keravec2026`	Full	IOPscience, open access. The primary cross-typology anchor and the empirical basis for treating climate suitability as a first-order adjustment.
`ziter2019`	Full	PubMed record, PMID 30910972.

continued on next page

Table 20 continued

Key	Verified	Basis and notes
`zolch2016`	Full	Lund University research portal record. *Note:* PET is a thermal-comfort index, not air temperature; used for relative ranking of intervention families and for the green-roof street-level caveat, never as an air-temperature value.
`santamouris2014`	Full	Open PDF via EEA Climate-ADAPT; abstract read directly. *Critical caveat:* the 0.3–3 K figure describes city-scale mass deployment in simulation studies, not one green roof on one building, and is therefore not used as a site-level typology value.
`reid2009`	Full	PMC open access record, PMC2801183. Grounds the vulnerability sub-indicators and the treatment of low cooling access as an independent vulnerability dimension.
`ember`	Full	Licence and coverage confirmed on Ember's data pages; 215 countries, CC-BY-4.0, redistribution permitted with attribution.
`nardo2008`	Full	Full bibliographic record from the JRC repository; step sequence confirmed from OECD/JRC descriptions.
`worldbank2021`	Full	World Bank plain-text document retrieved directly. The documented justification for shipping no global default unit costs.
`bowler2010`	Metadata + finding (secondary)	Metadata from the Bangor University repository; the 0.94 °C finding from the publisher abstract description. Elsevier full text blocked.

continued on next page

Table 20 continued

Key	Verified	Basis and notes
`gunawardena2017`	Metadata + finding (secondary)	Metadata from TU Delft and Bath research portals; findings from the publisher abstract description. Elsevier full text blocked. Justifies the humidity caveat applied to blue typologies in tropical-wet climates.
`venter2021`	Metadata + finding (secondary)	Metadata from the ADS record and an author-hosted PDF; the surface-versus-canopy comparison from an indexed description. Publisher full text blocked. Basis for treating LST as a relative screening proxy only.
`akbari2001`	Metadata + finding (secondary)	Multiple indexing records and an LBNL institutional listing. Basis for *deriving* cooling-energy savings from estimated air temperature reduction, replacing unsourced per-typology energy factors.
`norton2015`	Metadata + framework description (secondary)	Metadata from the Monash University research portal; framework description secondary. The closest published antecedent to this tool. No numeric value is taken from it.
`rahman2020`	Metadata only	Publisher listing. Used qualitatively — to justify that canopy condition modulates performance — never for a numeric value.
`oke1976`	Metadata only	Widely indexed. Cited for the conceptual distinction between canopy-layer, boundary-layer, and surface urban heat islands, not for a numeric value.

Which tool values depend on non-full verification

Three dependencies are worth naming explicitly.

The **energy derivation** of Section 7.8 rests entirely on `akbari2001`, verified at *metadata + finding (secondary)*. The 2–4 % per °C sensitivity is the sole input to every energy and greenhouse-gas figure the tool produces.

The **pocket park and park upgrade envelopes** depend on `bowler2010`'s 0.94 °C mean, verified at the same level, which is also the empirical anchor for the modelling-bias correction of Section 6.5.1.

The **LST proxy restriction** of Section 5.5 rests on `venter2021`, again at the same level — though here the tool's use is qualitative, and the conclusion is independently supported by the conceptual distinction in `oke1976`.

In each case the tool's confidence ratings and stated limitations reflect the verification level, and in no case does a value verified only at *metadata only* carry a number.